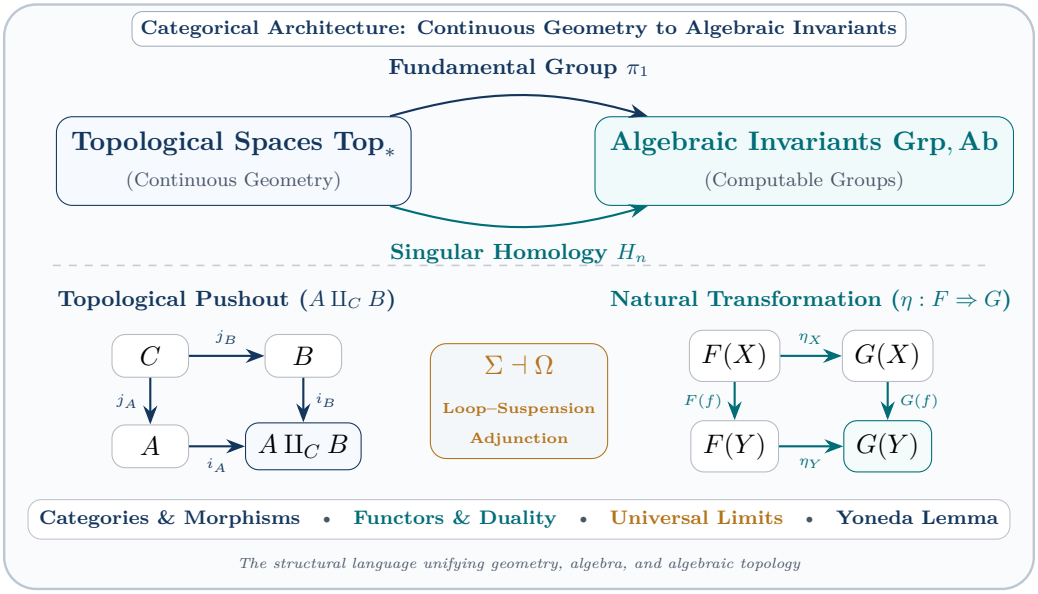
FOUNDATIONAL MONOGRAPHS IN PURE MATHEMATICS & TOPOLOGY

Introduction to Category Theory

A Treatise of the Abstract World: Foundational Bridges to Algebraic Topology



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An Advanced Pedagogical Treatise for Students of Pure Mathematics

Comprehensive Monograph on Arrow-Theoretic Foundations

Systematic Treatise Featuring Complete Deductions, Commutative Diagrams, and Topological Invariants

Undergraduate Texts in Pure Mathematics and Algebraic Topology

Preface

When an undergraduate student first opens a textbook on Algebraic Topology—such as Allen Hatcher's *Algebraic Topology* [6] or J. Peter May's *A Concise Course in Algebraic Topology* [7]—a striking shift in mathematical viewpoint takes place. In earlier courses on real analysis, linear algebra, or elementary group theory, one typically studies the internal anatomy of a *single* mathematical object at a time: the elements of a group G, the open subsets of a topological space X, or the eigenvalues of a linear operator T.

In Algebraic Topology, however, our central task is to build *bridges* between two seemingly different worlds: the continuous, flexible world of topological spaces and the discrete, rigid world of groups, rings, and modules. We do not merely assign a group $\pi_1(X, x_0)$ or $H_n(X)$ to a space X; we simultaneously translate every continuous map $f : X \to Y$ into a group homomorphism $f_* : H_n(X) \to H_n(Y)$. It was precisely to formalize this phenomenon that Samuel Eilenberg and Saunders Mac Lane invented **Category Theory** in their landmark 1945 paper *"General Theory of Natural Equivalences"* [1].

Pedagogical Design of This Treatise. Beginners often find category theory intimidating because of its high level of abstraction—famously nicknamed "generalized abstract nonsense" by Norman Steenrod (a phrase used affectionately by working mathematicians [13, 3]). To make this abstract world tangible and intuitive for an undergraduate at the very beginning of their mathematical career, these lecture notes follow four strict pedagogical principles:

1. **Topology-First Motivation:** Every abstract concept (categories, functors, natural transformations, limits, colimits, adjunctions, and the Yoneda Lemma) is immediately grounded in concrete examples from undergraduate algebra and algebraic topology [4, 5].

2. **Visual Schematic Diagrams at Every Stage:** Every single Axiom, Definition, Proposition, Theorem, Corollary, Remark, and Worked Solution includes a captioned commutative or schematic diagram, followed immediately by a shaded *Diagram Explanation* box that walks through how to read the arrows and paths.

3. **Worked Problems with Complete Deductions:** Each section features explicit Example Problems framed with a clear *Question* and a rigorous, line-by-line *Detailed Solution*, followed by closely aligned *Practice Exercises* to build fluency.

4. **Rigorous Cross-Referencing & Literature Grounding:** All definitions, axioms, and propositions are cross-referenced throughout the proofs, and citations connect you directly to accessible open-access resources on Wikipedia [13], arXiv [4, 9], ResearchGate [10, 11, 12], and classic textbooks [2, 3, 5, 8].

How to Read These Notes. Keep a pencil and scratch paper beside you. Whenever you encounter a commutative diagram, trace each path of arrows from the starting object (source) to the ending object (target) and verify why the two composite journeys agree. By the end of this treatise, diagram chasing will feel as natural as solving a system of linear equations.

Contents

Preface i

Notations Frequently Used iv

1 Categories, Axioms, and the Philosophy of Arrows 1

1.1 The Formal Definition and Structural Architecture of a Category . . . 1

1.2 The Landscape of Categories: From Algebra to Algebraic Topology . . . 4

1.3 Worked Example Problems and Practice Exercises for Section 1 . . . 5

2 Morphisms, Isomorphisms, Monomorphisms, and Epimorphisms 7

2.1 Isomorphisms, Sections, and Retractions . . . 7

2.2 Monomorphisms and Epimorphisms . . . 9

2.3 Worked Example Problem and Practice Exercises for Section 2 . . . 10

3 Functors: Structure-Preserving Bridges Between Worlds 11

3.1 Covariant and Contravariant Functors . . . 12

3.2 Key Functors in Algebra and Algebraic Topology . . . 13

3.2.1 Forgetful Functors: Stripping Structure . . . 13

3.2.2 The Fundamental Group Functor $\pi_1 : \mathbf{Top}_* \to \mathbf{Grp}$ . . . 14

3.2.3 Singular Homology H_n and Cohomology H^n: Covariant vs. Contravariant Duality . . . 15

3.2.4 Representable Hom-Functors: The Category's Internal Radar . . . 16

3.3 The Functorial Pipeline: Homotopy Invariance and Category Factoring . . . 17

3.3.1 Morphism Reflection and Faithful Functors . . . 17

3.4 Worked Example Problem and Practice Exercises for Section 3 . . . 19

4 Natural Transformations and Equivalence of Categories 20

4.1 Natural Transformations and Natural Isomorphisms . . . 20

4.2 Equivalence of Categories . . . 22

4.3 Worked Example Problems and Practice Exercises for Section 4 . . . 22

5 Universal Properties, Limits, and Colimits 24

5.1 Initial and Terminal Objects . . . 24

5.2 Products and Coproducts . . . 25

5.3 Pullbacks, Pushouts, and the Seifert–van Kampen Theorem . . . 26

5.4 Worked Example Problems and Practice Exercises for Section 5 . . . 27

6 Adjunctions, Representable Functors, and the Yoneda Lemma 29

6.1 Adjoint Functors . . . 29

6.2 The Yoneda Lemma: The Fundamental Theorem of Category Theory . . . 31

6.3 Worked Example Problems and Practice Exercises for Section 6 . . . 32

7 Where Will You Apply These Concepts? A Roadmap for Algebraic Topology and Beyond **33**

A Set-Theoretic Foundations: Small vs. Locally Small Categories **36**

B Abstract Algebra Foundations: The Algebraic Spine of Topology **36**

B.1 Groups, Homomorphisms, and the First Isomorphism Theorem 36

B.2 Free Groups, Presentations, and Amalgamated Pushouts 37

B.3 Cayley's Representation Theorem as the Prototypical Yoneda Lemma 38

B.4 Rings, Ideals, and Coefficient Fields in Cohomology 38

C Compendium of Canonical Functors and Functoriality Verification **39**

D Hints and Solution Outlines for Selected Practice Exercises **41**

References and Further Reading **43**

Notations Frequently Used

Notation	Mathematical Meaning and Standard Interpretation
1. Abstract Categories, Objects, and Morphisms	
$\mathcal{C}, \mathcal{D}, \mathcal{E}$	Arbitrary abstract categories (typically calligraphic font).
$\mathrm{Ob}(\mathcal{C})$	The class (or collection) of all **objects** in category $\mathcal{C}$.
$\mathrm{Hom}_{\mathcal{C}}(A, B)$ or $\mathcal{C}(A, B)$	The collection (hom-set in locally small categories) of all **morphisms** from A to B.
$f : A \to B$	A morphism $f \in \mathrm{Hom}_{\mathcal{C}}(A, B)$ with **domain** (source) A and **codomain** (target) B.
$g \circ f$ or gf	The composite morphism $A \xrightarrow{f} B \xrightarrow{g} C$ in $\mathcal{C}$.
id_A or 1_A	The **identity morphism** $A \to A$ associated with object $A \in \mathrm{Ob}(\mathcal{C})$.
$\mathcal{C}^{\mathrm{op}}$	The **opposite (or dual) category** formed by reversing all arrow directions in $\mathcal{C}$.
$f : A \xrightarrow{\sim} B$ or $A \cong B$	An **isomorphism** between objects A and B (possesses a two-sided inverse f^{-1}).
$f : A \hookrightarrow B$	A **monomorphism** (left-cancellable morphism: $f \circ u = f \circ v \implies u = v$).
$f : A \twoheadrightarrow B$	An **epimorphism** (right-cancellable morphism: $u \circ f = v \circ f \implies u = v$).
2. Concrete Mathematical Categories	
Set	Category of sets (objects: sets; morphisms: functions).
Grp, **Ab**	Categories of groups and abelian groups (morphisms: group homomorphisms).
Ring, R-**Mod**, $\mathbf{Vect}_k$	Categories of unital rings, left R-modules, and vector spaces over a field k.
Top	Category of topological spaces (objects: spaces; morphisms: continuous maps).
$\mathbf{Top}_*$	Category of pointed topological spaces (X, x_0) and basepoint-preserving continuous maps.
hTop, $\mathbf{hTop}_*$	Homotopy categories of spaces (morphisms: homotopy classes $[f]$ of continuous maps).
$\mathbf{Ch}(R)$ or $\mathbf{Ch}_\bullet(\mathbf{Ab})$	Category of chain complexes of R-modules or abelian groups (morphisms: chain maps).
3. Functors and Natural Transformations	
$F : \mathcal{C} \to \mathcal{D}$	A **covariant functor** (preserves morphism directions and composition).
$F : \mathcal{C}^{\mathrm{op}} \to \mathcal{D}$	A **contravariant functor** (reverses arrow directions: $F(g \circ f) = F(f) \circ F(g)$).
$\eta : F \Rightarrow G$	A **natural transformation** with components $\eta_X : F(X) \to G(X)$ satisfying naturality.

Notation	Mathematical Meaning and Standard Interpretation *(Continued)*
$[\mathcal{C}, \mathcal{D}]$ or $\mathbf{Fun}(\mathcal{C}, \mathcal{D})$	The **functor category** (objects: functors $\mathcal{C} \to \mathcal{D}$; morphisms: natural transformations).
$F \dashv G$	An **adjunction** where F is left adjoint to G ($\mathrm{Hom}(F(X), Y) \cong \mathrm{Hom}(X, G(Y))$).
4. Universal Constructions, Limits, and Colimits	
$\varnothing, *$	An **initial object** ($\varnothing$) and a **terminal object** ($*$) in category $\mathcal{C}$.
$A \times B$, $\prod_{i \in I} A_i$	Categorical **product** equipped with canonical projections π_1, π_2.
$A \amalg B$ or $A \sqcup B$	Categorical **coproduct** (disjoint union in **Top**, free product $G * H$ in **Grp**).
$A \times_C B$	**Pullback** (fiber product) over C of cospan $A \to C \leftarrow B$.
$A \amalg_C B$	**Pushout** (amalgamated sum) under C of span $A \leftarrow C \to B$.
5. Algebraic Topology and Homotopy Invariants	
$\pi_1 : \mathbf{Top}_* \to \mathbf{Grp}$	The **fundamental group functor** assigning $(X, x_0) \mapsto \pi_1(X, x_0)$.
$H_n : \mathbf{Top} \to \mathbf{Ab}$	The n-th **singular homology functor** (covariant).
$H^n(-; G) : \mathbf{Top}^{\mathrm{op}} \to \mathbf{Ab}$	The n-th **singular cohomology functor** with coefficients in G (contravariant).
$X \vee Y$	**Wedge sum** of pointed spaces (coproduct in $\mathbf{Top}_*$: $(X \sqcup Y)/(x_0 \sim y_0)$).
$\Sigma \dashv \Omega$	The **loop-suspension adjunction** in $\mathbf{Top}_*$ / $\mathbf{hTop}_*$ ($[\Sigma X, Y]_* \cong [X, \Omega Y]_*$).

1 Categories, Axioms, and the Philosophy of Arrows

In classical set-theoretic mathematics, we define a structure by specifying a set of elements and operations on those elements. Category theory shifts the spotlight: *we understand a mathematical object not by looking inside at its elements, but by observing how it interacts with all other objects of the same kind via structure-preserving maps (arrows)* [14, 4, 3].

1.1 The Formal Definition and Structural Architecture of a Category

In classical abstract algebra, an algebraic system (such as a group, ring, or vector space) is defined by a set of elements equipped with globally defined operations. Category theory shifts this paradigm: it organizes every mathematical universe into a two-tiered architecture consisting of **Constituent Data** (the underlying directed multigraph of objects, arrows, identities, and composition) governed by **Coherence Axioms** (the laws of associativity and unitality).

Definition 1.1 (Category [2, 5, 14])**.** A **category** $\mathcal{C}$ consists of four constituent pieces of algebraic data satisfying two foundational coherence axioms:

I. Constituent Data (The Algebraic Skeleton):

1. **A Collection of Objects,** denoted $\mathrm{Ob}(\mathcal{C})$, whose elements represent the mathematical structures of interest (e.g., sets in **Set**, groups in **Grp**, or topological spaces in **Top**). Objects are typically symbolized by capital letters $A, B, C, X, Y, \ldots$

2. **Collections of Morphisms,** denoted $\mathrm{Hom}_{\mathcal{C}}(A, B)$ (or $\mathcal{C}(A, B)$) for every ordered pair of objects $(A, B) \in \mathrm{Ob}(\mathcal{C}) \times \mathrm{Ob}(\mathcal{C})$. An element $f \in \mathrm{Hom}_{\mathcal{C}}(A, B)$ is called a **morphism** (or **arrow**) from source A to target B, denoted diagrammatically by:

$$f : A \longrightarrow B \quad \text{or} \quad A \xrightarrow{f} B.$$

   We write $A = \mathrm{dom}(f)$ for the **domain** (source) of f, and $B = \mathrm{cod}(f)$ for the **codomain** (target) of f. Unlike simple directed graphs where at most one edge connects two vertices, a category is a *directed multigraph* (or quiver): multiple distinct morphisms $f_1 \neq f_2 : A \rightrightarrows B$ can share the exact same domain and codomain (see Figure 1).

3. **An Identity Assignment,** providing for every object $A \in \mathrm{Ob}(\mathcal{C})$ a distinguished morphism:

$$\mathrm{id}_A \in \mathrm{Hom}_{\mathcal{C}}(A, A) \quad (\text{also denoted } 1_A),$$

   called the **identity morphism** on A. Geometrically, id_A is an endomorphism loop based at A. Under composition, it acts as a two-sided neutral element (governed by Axiom 1.3), and Proposition 1.4 proves that this identity arrow is strictly unique for each object.

4. **A Composition Law,** consisting of a family of binary operations:

$$\circ : \mathrm{Hom}_{\mathcal{C}}(B, C) \times \mathrm{Hom}_{\mathcal{C}}(A, B) \longrightarrow \mathrm{Hom}_{\mathcal{C}}(A, C), \qquad (g, f) \longmapsto g \circ f,$$

   defined for every ordered triple (A, B, C) of objects. Crucially, composition is a **partial binary operation** across the category: the composite $g \circ f$ exists *if and only if* the codomain of the first morphism matches the domain of the second morphism:

$$\mathrm{cod}(f) = \mathrm{dom}(g).$$

   Following the standard convention of functions, $g \circ f$ denotes "apply f first, then apply g". Diagrammatically, any such composable sequence forms the legs of a commutative triangle completed by the composite hypotenuse $g \circ f : A \to C$ (see Figure 2), with associativity governed by Axiom 1.2.

II. Coherence Axioms (The Governing Laws):

1. **Associativity of Composition (Axiom 1.2):** For any composable sequence of morphisms $A \xrightarrow{f} B \xrightarrow{g} C \xrightarrow{h} D$,
$$h \circ (g \circ f) = (h \circ g) \circ f.$$

2. **Left and Right Unitality (Axiom 1.3):** For every morphism $f : A \to B$,
$$\mathrm{id}_B \circ f = f = f \circ \mathrm{id}_A.$$

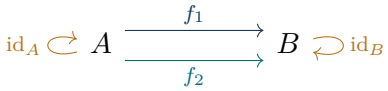


Figure 1: Quiver anatomy of a category: objects A, B, distinct parallel morphisms ($f_1 \neq f_2$), and identity loops ($\mathrm{id}_A, \mathrm{id}_B$).

Diagram Explanation: In Figure 1, the structural data of objects and arrows are displayed:

- **Parallel Morphisms:** Unlike simple relational structures, the hom-set $\mathrm{Hom}_{\mathcal{C}}(A, B)$ can contain multiple distinct arrows (such as $f_1 \neq f_2$) connecting the exact same source A to the same target B (e.g., different continuous maps between topological spaces, or different matrices between vector spaces).
- **Identity Endomorphisms:** Every object A is equipped with a distinguished self-arrow $\mathrm{id}_A : A \to A$ (gold loop) that leaves morphisms unchanged under composition (Axiom 1.3 and Proposition 1.4).

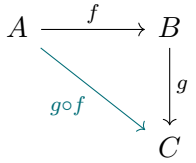


Figure 2: The composition law: the composable sequence $A \xrightarrow{f} B \xrightarrow{g} C$ and the commutative composite triangle ($g \circ f$).

Diagram Explanation: Figure 2 defines the geometric meaning of a **commutative diagram**:

- **Matching Condition:** Composition is defined strictly because $\mathrm{cod}(f) = B = \mathrm{dom}(g)$. The composite arrow $g \circ f : A \to C$ (teal hypotenuse) bypasses the intermediate object B.
- **Path Invariance:** In category theory, stating that this triangle **commutes** does *not* mean arrows commute like numbers ($g \circ f \neq f \circ g$). It means that **different directed paths between the same initial object (A) and final object (C) yield the exact same morphism**: traveling along f and then g is algebraically identical to traveling directly along $g \circ f$.

Mathematical Concept	Standard Notation	Definition and Structural Role
Hom-Set / Hom-Class	$\mathrm{Hom}_{\mathcal{C}}(A, B)$ or $\mathcal{C}(A, B)$	Collection of all directed morphisms from domain A to codomain B.
Endomorphism	$f \in \mathrm{End}_{\mathcal{C}}(A)$	Any morphism whose source and target coincide ($f : A \to A$).
Endomorphism Monoid	$(\mathrm{End}_{\mathcal{C}}(A), \circ, \mathrm{id}_A)$	The set $\mathrm{Hom}_{\mathcal{C}}(A, A)$ forms a monoid under associative composition (Axiom 1.2) with neutral element id_A (Axiom 1.3).
Automorphism Group	$\mathrm{Aut}_{\mathcal{C}}(A)$	The group of all invertible endomorphisms on A under composition (e.g., homeomorphisms $X \xrightarrow{\sim} X$ in **Top**).

Table 2: Taxonomy of morphism collections associated with objects in an arbitrary category $\mathcal{C}$.

Why Partial Composition Requires Arrow-Chaining Associativity: In classical algebra, an operation like $+$ or $\times$ combines any two elements unconditionally. In category theory, morphisms represent *directed processes or pathways*. Composition $g \circ f$ is an operational relay: you can only pass the baton to g if you have arrived at the destination of f. Associativity does not assert that all paths commute; it guarantees that *macro-pathways are independent of temporal grouping*. Whether an algebraic topologist decomposes a continuous path into two segments followed by a third, or a first segment followed by two, the resulting composite traversal across the space remains identical.

Axiom 1.2 (Associativity of Composition [2, 4])**.** For any four objects $A, B, C, D \in \mathrm{Ob}(\mathcal{C})$ and any three composable morphisms $f : A \to B$, $g : B \to C$, and $h : C \to D$, composition is associative:

$$h \circ (g \circ f) = (h \circ g) \circ f \quad \text{in } \mathrm{Hom}_{\mathcal{C}}(A, D). \tag{1}$$

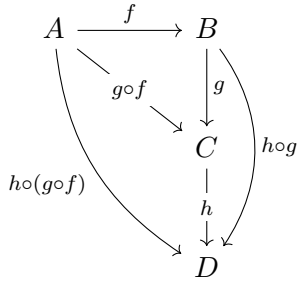


Figure 3: Commutative diagram illustrating Axiom 1.2 (Associativity of Composition).

Diagram Explanation: Figure 3 displays two ways to collapse the three-step path $A \xrightarrow{f} B \xrightarrow{g} C \xrightarrow{h} D$ into a single arrow $A \to D$. Either we first form the intermediate shortcut $g \circ f : A \to C$ and then post-compose with h (left curved arrow), or we first form $h \circ g : B \to D$ and pre-compose with f (right curved arrow). Axiom 1.2 asserts that both composite paths yield the identical morphism $A \to D$, allowing us to write $h \circ g \circ f$ unambiguously.

Axiom 1.3 (Left and Right Identity Laws [2, 3])**.** For every morphism $f : A \to B$ in $\mathcal{C}$, the identity morphisms $\mathrm{id}_A : A \to A$ and $\mathrm{id}_B : B \to B$ satisfy:

$$\mathrm{id}_B \circ f = f = f \circ \mathrm{id}_A. \tag{2}$$

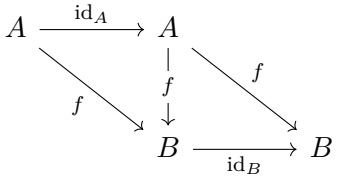


Figure 4: Commutative double-triangle illustrating Axiom 1.3 (Left and Right Identity Laws).

Diagram Explanation: In Figure 4, the left triangle states that pausing at A via id_A before applying f equals f ($f \circ \mathrm{id}_A = f$). The right triangle states that applying f first and then pausing at B via id_B also equals f ($\mathrm{id}_B \circ f = f$). Thus identity morphisms behave like the number 1 in multiplication.

From Existence to Uniqueness (Miracle of the Coherence Axioms): In Definition 1.1(3), we only postulated the *existence* of an identity arrow id_A for each object. Just as Lemma 3.15 in MA204 proves that a group's identity element is unique by pitting two candidate identities against each other ($e = e * f = f$), our categorical coherence axioms automatically force the identity arrow to be strictly unique.

Proposition 1.4 (Uniqueness of the Identity Morphism)**.** *For every object A in a category $\mathcal{C}$, the identity morphism $\mathrm{id}_A \in \mathrm{Hom}_{\mathcal{C}}(A, A)$ is unique.*

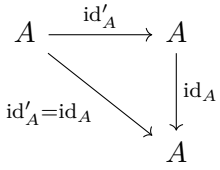


Figure 5: Composing two candidate identities id_A and id'_A on the same object A.

Diagram Explanation: Figure 5 shows how to compare two candidate identity arrows id_A and id'_A in $\mathrm{Hom}_{\mathcal{C}}(A, A)$ by composing them directly with one another.

Proof. Suppose id_A and id'_A are both identity morphisms on A satisfying Axiom 1.3. Treating id_A as a left identity and id'_A as an arbitrary arrow $A \to A$, Equation (2) gives $\mathrm{id}_A \circ \mathrm{id}'_A = \mathrm{id}'_A$. Conversely, treating id'_A as a right identity and id_A as an arbitrary arrow $A \to A$, Equation (2) gives $\mathrm{id}_A \circ \mathrm{id}'_A = \mathrm{id}_A$. Combining both equalities yields $\mathrm{id}_A = \mathrm{id}_A \circ \mathrm{id}'_A = \mathrm{id}'_A$. □

1.2 The Landscape of Categories: From Algebra to Algebraic Topology

To appreciate the unifying power of Definition 1.1, let us catalog the foundational categories you will encounter constantly in Algebraic Topology [6, 7, 10]:

- **Concrete Algebraic Categories: Set** (sets and functions), **Grp** (groups and group homomorphisms), **Ab** (abelian groups and group homomorphisms), **Ring** (rings and ring homomorphisms), and $\mathbf{Vect}_k$ (vector spaces over a field k and k-linear transformations).
- **Topological Categories (Top and $\mathbf{Top}_*$):** In **Top**, objects are topological spaces (X, τ_X) and morphisms are continuous maps $f : X \to Y$. In the **pointed topological category $\mathbf{Top}_*$**, objects are pairs (X, x_0) where $x_0 \in X$ is a chosen basepoint, and morphisms $f : (X, x_0) \to (Y, y_0)$ are continuous maps satisfying $f(x_0) = y_0$.
- **The Homotopy Category (hTop and $\mathbf{hTop}_*$):** In Algebraic Topology, two continuous maps $f, g : X \to Y$ are *homotopic* ($f \simeq g$) if there is a continuous family of maps $H : X \times [0, 1] \to Y$ deforming f into g. As shown in Worked Example 1.6, taking $\mathrm{Ob}(\mathbf{hTop}) = \mathrm{Ob}(\mathbf{Top})$ and defining $\mathrm{Hom}_{\mathbf{hTop}}(X, Y) = [X, Y]$ (the set of homotopy equivalence classes $[f]$ of continuous maps) yields a genuine category where arrows are no longer raw functions, but equivalence classes of functions!
- **Categories Without Internal "Elements" (Monoids and Posets):**

- A **monoid** $(M, \cdot, e)$ is a category $\mathcal{B}M$ with *strictly one object* $\bullet$, where $\mathrm{Hom}_{\mathcal{B}M}(\bullet, \bullet) = M$ and composition of arrows is monoid multiplication.
- A **partially ordered set (poset)** $(P, \leq)$ is a category where $\mathrm{Ob}(\mathcal{P}) = P$, and for any $x, y \in P$, the hom-set $\mathrm{Hom}_{\mathcal{P}}(x, y)$ contains a single arrow $x \to y$ if and only if $x \leq y$, and is empty otherwise. Reflexivity ($x \leq x$) gives identities, and transitivity ($x \leq y$ and $y \leq z \implies x \leq z$) gives composition!

The Principle of Categorical Duality (Two Theorems for the Price of One): In set theory, you cannot simply reverse a function $f : A \to B$ unless f is bijective. In category theory, however, arrows are primitive symbols governed solely by composition laws. If we formally reverse the source and target of every arrow in $\mathcal{C}$, we obtain a valid new category $\mathcal{C}^{\mathrm{op}}$ where a "morphism $A \to B$ in $\mathcal{C}^{\mathrm{op}}$" is simply a morphism $B \to A$ in $\mathcal{C}$. This formal symmetry is the secret behind why every concept in topology has a twin: homology vs. cohomology, products vs. coproducts, and projections vs. inclusions.

Remark 1.5 (Opposite Categories and the Principle of Duality [2, 3]). Given any category $\mathcal{C}$, we construct its **opposite (dual) category** $\mathcal{C}^{\mathrm{op}}$ by keeping the exact same objects $\mathrm{Ob}(\mathcal{C}^{\mathrm{op}}) = \mathrm{Ob}(\mathcal{C})$, but reversing every arrow:

$$\mathrm{Hom}_{\mathcal{C}^{\mathrm{op}}}(A, B) := \mathrm{Hom}_{\mathcal{C}}(B, A).$$

If $f^{\mathrm{op}} \in \mathrm{Hom}_{\mathcal{C}^{\mathrm{op}}}(A, B)$ and $g^{\mathrm{op}} \in \mathrm{Hom}_{\mathcal{C}^{\mathrm{op}}}(B, C)$, their composite in $\mathcal{C}^{\mathrm{op}}$ is defined by $g^{\mathrm{op}} \circ_{\mathrm{op}} f^{\mathrm{op}} := (f \circ_{\mathcal{C}} g)^{\mathrm{op}}$. Because Axiom 1.2 and Axiom 1.3 are completely symmetric under arrow reversal, *every theorem proved from the axioms of a general category $\mathcal{C}$ automatically yields a dual theorem in $\mathcal{C}^{\mathrm{op}}$ simply by reversing all arrows!*

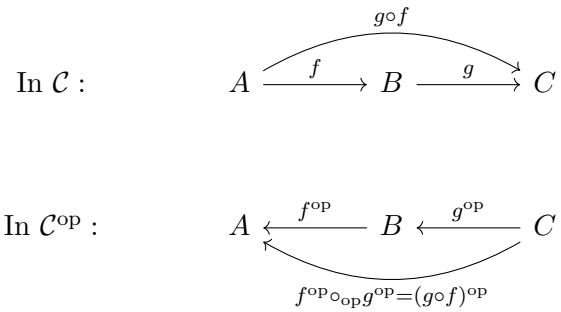


Figure 6: Duality between composition in $\mathcal{C}$ (top) and reversed composition in $\mathcal{C}^{\mathrm{op}}$ (bottom).

Diagram Explanation: In Figure 6, reversing the direction of arrows $f : A \to B$ and $g : B \to C$ in $\mathcal{C}$ turns them into arrows $g^{\mathrm{op}} : C \to B$ and $f^{\mathrm{op}} : B \to A$ in $\mathcal{C}^{\mathrm{op}}$. Notice how the order of composition reverses: in $\mathcal{C}^{\mathrm{op}}$, we first travel from C to B via g^{op} and then from B to A via f^{op}.

1.3 Worked Example Problems and Practice Exercises for Section 1

Worked Example 1.6 (Verification of the Homotopy Category **hTop** [6, 7])**. Question:** Let $\mathrm{Ob}(\mathbf{hTop})$ be the collection of all topological spaces. For any two spaces X, Y, let $\mathrm{Hom}_{\mathbf{hTop}}(X, Y) = [X, Y]$ be the set of homotopy classes $[f]$ of continuous maps $f : X \to Y$. Define composition by $[g] \circ [f] := [g \circ f]$. Prove carefully that composition is well-defined and that **hTop** satisfies Definition 1.1.

Detailed Solution and Deduction:

1. **Well-definedness of composition:** Suppose $f_0, f_1 : X \to Y$ are homotopic via a continuous homotopy $F : X \times [0, 1] \to Y$ (so $F(x, 0) = f_0(x)$ and $F(x, 1) = f_1(x)$), and $g_0, g_1 : Y \to Z$ are

homotopic via $G : Y \times [0,1] \to Z$. We must show that $[g_0 \circ f_0] = [g_1 \circ f_1]$ in $[X, Z]$. Define two intermediate homotopies $H_1, H_2 : X \times [0,1] \to Z$ by:

$$H_1(x,t) = g_0(F(x,t)), \qquad H_2(x,t) = G(f_1(x), t).$$

Since g_0, F, G, f_1 are continuous and compositions of continuous maps in **Top** are continuous, H_1 is a homotopy from $g_0 \circ f_0$ to $g_0 \circ f_1$, and H_2 is a homotopy from $g_0 \circ f_1$ to $g_1 \circ f_1$. By transitivity of homotopy (pasting H_1 on $[0, 1/2]$ and H_2 on $[1/2, 1]$), $g_0 \circ f_0 \simeq g_1 \circ f_1$, so $[g_0 \circ f_0] = [g_1 \circ f_1]$.

2. **Verification of Axiom 1.2 and Axiom 1.3:** For composable classes $[f] \in [X, Y]$, $[g] \in [Y, Z]$, and $[h] \in [Z, W]$, function composition in **Top** is strictly associative, so:

$$[h] \circ ([g] \circ [f]) = [h \circ (g \circ f)] = [(h \circ g) \circ f] = ([h] \circ [g]) \circ [f].$$

Similarly, setting $\mathrm{id}_X := [1_X]$ (the homotopy class of the identity map $1_X : X \to X$), we have $[1_Y] \circ [f] = [1_Y \circ f] = [f] = [f \circ 1_X] = [f] \circ [1_X]$. Thus **hTop** is a category. □

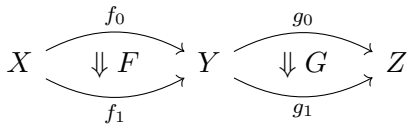


Figure 7: Horizontal composition of homotopies $F : f_0 \simeq f_1$ and $G : g_0 \simeq g_1$ in **hTop**.

Diagram Explanation: In Figure 7, the double arrows $\Downarrow F$ and $\Downarrow G$ represent continuous deformations (homotopies) between the upper and lower continuous maps. In the category **hTop**, the entire lens between f_0 and f_1 collapses into a single morphism $[f_0] = [f_1] : X \to Y$.

Worked Example 1.7 (The Fundamental Groupoid $\Pi_1(X)$ as a Category [6, 20])**. Question:** Let X be a topological space. Construct a category $\Pi_1(X)$ whose objects are the points of X, and whose morphisms from x to y are homotopy classes $[\gamma]$ of paths $\gamma : [0,1] \to X$ with $\gamma(0) = x$ and $\gamma(1) = y$ (relative to endpoints). Explain how the fundamental group $\pi_1(X, x_0)$ appears inside $\Pi_1(X)$.

Detailed Solution and Deduction: Let $\mathrm{Ob}(\Pi_1(X)) = X$. For $x, y \in X$, let $\mathrm{Hom}_{\Pi_1(X)}(x, y)$ be the set of path-homotopy classes $[\gamma]$ of continuous paths from x to y. Given $[\gamma_1] \in \mathrm{Hom}_{\Pi_1(X)}(x, y)$ and $[\gamma_2] \in \mathrm{Hom}_{\Pi_1(X)}(y, z)$, define categorical composition by path concatenation:

$$[\gamma_2] \circ [\gamma_1] := [\gamma_1 * \gamma_2], \quad \text{where } (\gamma_1 * \gamma_2)(s) = \begin{cases} \gamma_1(2s) & 0 \le s \le 1/2, \\ \gamma_2(2s-1) & 1/2 \le s \le 1. \end{cases}$$

Although raw path concatenation is not strictly associative, reparametrization up to homotopy ensures $[(\gamma_1 * \gamma_2) * \gamma_3] = [\gamma_1 * (\gamma_2 * \gamma_3)]$ (satisfying Axiom 1.2). The constant path $c_x(s) = x$ provides the identity $[\mathrm{id}_x] = [c_x]$ (Axiom 1.3), and the reversed path $\bar{\gamma}(s) = \gamma(1-s)$ makes every arrow invertible. Restricting $\Pi_1(X)$ to a single object $x_0 \in X$ yields the endomorphism group $\mathrm{Hom}_{\Pi_1(X)}(x_0, x_0) = \pi_1(X, x_0)$! □

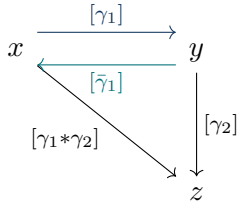


Figure 8: Path concatenation and path reversal in the fundamental groupoid $\Pi_1(X)$.

Diagram Explanation: In Figure 8, the components of the fundamental groupoid $\Pi_1(X)$ are represented geometrically:

- **Forward Morphism $[\gamma_1] : x \to y$:** The upper horizontal arrow (in **Navy**, pointing right $\to$) represents the homotopy class of the path γ_1 from source x to target y.
- **Inverse Morphism $[\bar{\gamma}_1] : y \to x$:** The lower horizontal arrow (in **Teal**, pointing left $\leftarrow$) represents the reversed path $\bar{\gamma}_1(s) = \gamma_1(1-s)$. Because the concatenated loop $[\bar{\gamma}_1] \circ [\gamma_1] = [\gamma_1 * \bar{\gamma}_1] = [\mathrm{id}_x] = [c_x]$ is path-homotopic to the stationary constant loop at x, $[\bar{\gamma}_1]$ is the two-sided categorical inverse $[\gamma_1]^{-1}$.
- **Categorical Composition $[\gamma_1 * \gamma_2] : x \to z$:** The diagonal hypotenuse represents the composite morphism $[\gamma_2] \circ [\gamma_1]$ obtained by traveling first along γ_1 to y and then along γ_2 to z (vertical arrow $\downarrow$).

Practice Exercise 1.1. Let $\mathcal{C}$ be a category and fix an object $B \in \mathrm{Ob}(\mathcal{C})$. Define the **slice category** $\mathcal{C}/B$ whose objects are pairs (X, p_X) where $p_X : X \to B$ is a morphism in $\mathcal{C}$, and where a morphism $f : (X, p_X) \to (Y, p_Y)$ is a morphism $f : X \to Y$ in $\mathcal{C}$ such that $p_Y \circ f = p_X$. Draw the commutative triangle for a morphism in $\mathcal{C}/B$ and prove that $\mathcal{C}/B$ satisfies Axiom 1.2 and Axiom 1.3. (Note: In Algebraic Topology, $\mathbf{Top}/B$ is the category of spaces over a base space B, central to covering spaces and fiber bundles!)

Practice Exercise 1.2. Given two categories $\mathcal{C}$ and $\mathcal{D}$, define the **product category** $\mathcal{C} \times \mathcal{D}$ with objects $(C, D) \in \mathrm{Ob}(\mathcal{C}) \times \mathrm{Ob}(\mathcal{D})$ and morphisms $(f, g) : (C, D) \to (C', D')$. Verify all category axioms.

Conceptual Bridge to Section 2: The Elimination of Elements

In undergraduate algebra (MA204) and analysis, we classify functions by inspecting individual elements: f is injective if $f(x) = f(y) \implies x = y$, and surjective if every element in the target has a preimage. But in abstract categories, *objects are opaque black boxes containing no elements.* How can we detect that a continuous map is an inclusion of a subspace, or that a quotient map collapses boundaries, without ever mentioning a single point x? In Section 2, we replace element-wise probing with **arrow-cancellation tests**, transforming the notions of injectivity and surjectivity into the categorical concepts of **monomorphisms** and **epimorphisms**.

2 Morphisms, Isomorphisms, Monomorphisms, and Epimorphisms

In **Set**, we classify functions as injective (one-to-one), surjective (onto), or bijective by talking about elements $x \in A$. How do we express these concepts in an arbitrary category $\mathcal{C}$ where objects have no internal elements? We translate element-wise conditions into *cancellation properties of arrows* [4, 5].

2.1 Isomorphisms, Sections, and Retractions

Definition 2.1 (Isomorphism, Section, and Retraction [2, 3])**.** Let $f : A \to B$ be a morphism in a category $\mathcal{C}$.

1. A morphism $g : B \to A$ is a **left inverse** (or **retraction**) of f if $g \circ f = \mathrm{id}_A$. In this case, f is called a **section** (or **split monomorphism**) of g.
2. A morphism $h : B \to A$ is a **right inverse** (or **section**) of f if $f \circ h = \mathrm{id}_B$. In this case, f is called a **retraction** (or **split epimorphism**) of h.
3. The morphism $f : A \to B$ is an **isomorphism** (written $f : A \xrightarrow{\sim} B$) if there exists a two-sided inverse $g : B \to A$ such that $g \circ f = \mathrm{id}_A$ and $f \circ g = \mathrm{id}_B$. If such a g exists, we write $g = f^{-1}$ and say A and B are **isomorphic** ($A \cong B$).

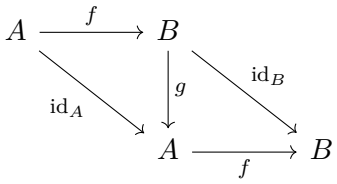


Figure 9: Commutative triangles defining a two-sided inverse $g = f^{-1}$ of an isomorphism $f : A \to B$.

Diagram Explanation: In Figure 9, the left triangle encodes $g \circ f = \mathrm{id}_A$ (g undoes f on A), while the right triangle encodes $f \circ g = \mathrm{id}_B$ (f undoes g on B). In **Set**, isomorphisms are bijections; in **Grp**, group isomorphisms; in **Top**, **homeomorphisms**; and in **hTop**, **homotopy equivalences**!

Proposition 2.2 (Uniqueness of Inverses)**.** *If a morphism $f : A \to B$ in $\mathcal{C}$ has both a left inverse $g : B \to A$ ($g \circ f = \mathrm{id}_A$) and a right inverse $h : B \to A$ ($f \circ h = \mathrm{id}_B$), then $g = h$. In particular, the two-sided inverse f^{-1} of an isomorphism is unique.*

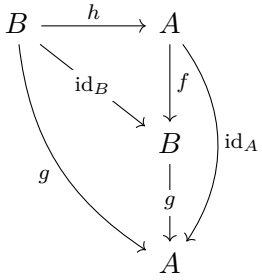


Figure 10: Pasting the left-inverse triangle ($g \circ f = \mathrm{id}_A$) and right-inverse triangle ($f \circ h = \mathrm{id}_B$).

Diagram Explanation: Figure 10 shows the classic associativity chain: starting at B, going along $h \to f \to g$ can be grouped either as $(g \circ f) \circ h = \mathrm{id}_A \circ h = h$ (right curved route) or as $g \circ (f \circ h) = g \circ \mathrm{id}_B = g$ (left curved route).

Proof. Using Axiom 1.3 and Axiom 1.2, we compute:

$$g = g \circ \mathrm{id}_B = g \circ (f \circ h) = (g \circ f) \circ h = \mathrm{id}_A \circ h = h. \qquad \square$$

Remark 2.3 (Topological Retractions in **Top** [6]). In Algebraic Topology, if $A \subseteq X$ is a subspace and $i : A \hookrightarrow X$ is the inclusion map, a continuous map $r : X \to A$ such that $r \circ i = \mathrm{id}_A$ (i.e., $r|_A = \mathrm{id}_A$) is called a **retraction** of X onto A, and A is called a **retract** of X. Notice how Definition 2.1(1) directly generalizes this geometric concept to any category!

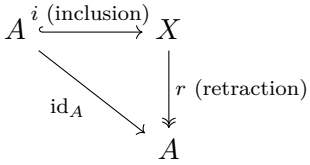


Figure 11: Geometric retraction of an ambient space X onto a subspace $A \subseteq X$ ($r \circ i = \mathrm{id}_A$).

Diagram Explanation: In Figure 11, the inclusion $i : A \hookrightarrow X$ is a section (split monomorphism) and the continuous collapse $r : X \twoheadrightarrow A$ is a retraction (split epimorphism). Every point of X is mapped continuously into A while points already in A remain fixed ($r(a) = a$).

From Geometric Retracts to General Subspaces: A split monomorphism (section) requires an explicit, globally defined retraction r that collapses the ambient space back onto the subspace

without tearing. In topology, this is an extraordinarily stringent demand: for example, the inclusion of the circle into the punctured plane $S^1 \hookrightarrow \mathbb{R}^2 \setminus \{0\}$ admits a retraction, but the boundary inclusion of the circle into the closed disk $S^1 \hookrightarrow D^2$ admits *no continuous retraction* (as we will prove via algebraic invariants in Worked Example 3.12). If we required every subspace inclusion to be split, most geometric embeddings would be discarded! We therefore weaken the demand: instead of requiring an inverse arrow to exist, we require only that f *preserves distinctions among incoming test arrows*. This yields the general concept of a **monomorphism**.

2.2 Monomorphisms and Epimorphisms

Definition 2.4 (Monomorphism and Epimorphism [2, 4])**.** Let $f : A \to B$ be a morphism in a category $\mathcal{C}$.

1. f is a **monomorphism** (or **mono**, written $f : A \hookrightarrow B$) if it is *left-cancellable*: for every object $Z \in \mathrm{Ob}(\mathcal{C})$ and every pair of parallel morphisms $g_1, g_2 : Z \to A$,
$$f \circ g_1 = f \circ g_2 \implies g_1 = g_2. \tag{3}$$
2. f is an **epimorphism** (or **epi**, written $f : A \twoheadrightarrow B$) if it is *right-cancellable*: for every object $W \in \mathrm{Ob}(\mathcal{C})$ and every pair of parallel morphisms $h_1, h_2 : B \to W$,
$$h_1 \circ f = h_2 \circ f \implies h_1 = h_2. \tag{4}$$

$$Z \underset{g_2}{\overset{g_1}{\rightrightarrows}} A \xhookrightarrow{f\ \text{(mono)}} B \qquad\qquad A \xrightarrow{f\ \text{(epi)}} B \underset{h_2}{\overset{h_1}{\rightrightarrows}} W$$

Figure 12: Left: Monomorphism ($f \circ g_1 = f \circ g_2 \implies g_1 = g_2$). Right: Epimorphism ($h_1 \circ f = h_2 \circ f \implies h_1 = h_2$).

Diagram Explanation: Figure 12 shows that a monomorphism $f : A \hookrightarrow B$ can cancel from the left of a parallel fork $Z \rightrightarrows A$, whereas an epimorphism $f : A \twoheadrightarrow B$ can cancel from the right of a parallel fork $B \rightrightarrows W$. Notice that f is an epimorphism in $\mathcal{C}$ if and only if f^{op} is a monomorphism in $\mathcal{C}^{\mathrm{op}}$ (Remark 1.5).

Proposition 2.5 (Stability of Monomorphisms and Epimorphisms under Composition)**.** *Let $f : A \to B$ and $g : B \to C$ be morphisms in $\mathcal{C}$.*

1. *If f and g are both monomorphisms, then $g \circ f : A \to C$ is a monomorphism.*
2. *Conversely, if $g \circ f$ is a monomorphism, then f is necessarily a monomorphism.*
3. *Dually, if f and g are epimorphisms, $g \circ f$ is an epimorphism; if $g \circ f$ is an epimorphism, then g is an epimorphism.*

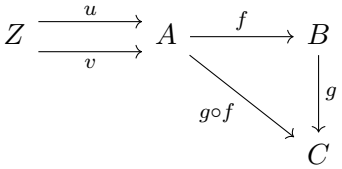


Figure 13: Testing left-cancellability of f and $g \circ f$ against a test pair $u, v : Z \rightrightarrows A$.

Diagram Explanation: In Figure 13, any two test morphisms $u, v : Z \rightrightarrows A$ that agree after f ($f \circ u = f \circ v$) must also agree after $g \circ f$. Thus if the diagonal arrow $g \circ f$ is left-cancellable, the first step f must also be left-cancellable.

Proof. (1) Suppose f and g are monomorphisms, and let $u, v : Z \rightrightarrows A$ satisfy $(g \circ f) \circ u = (g \circ f) \circ v$. By Axiom 1.2, $g \circ (f \circ u) = g \circ (f \circ v)$. Since g is mono, $f \circ u = f \circ v$. Since f is mono, $u = v$. (2) Suppose $g \circ f$ is mono and $f \circ u = f \circ v$. Applying g on the left gives $g \circ (f \circ u) = g \circ (f \circ v)$, so $(g \circ f) \circ u = (g \circ f) \circ v$. Since $g \circ f$ is mono, $u = v$. Statement (3) follows immediately by applying (1) and (2) in $\mathcal{C}^{\mathrm{op}}$. □

The Strict Hierarchy of Cancellation Laws: Proposition 2.5(2) yields an immediate structural dividend: if a morphism $f : A \to B$ admits a left inverse g ($g \circ f = \mathrm{id}_A$), then because id_A is a monomorphism, f is automatically forced to be a monomorphism. This establishes a strict three-tier hierarchy in every category:

$$\text{Isomorphism} \implies \text{Section (Split Mono)} \implies \text{Monomorphism},$$

and dually Isomorphism $\implies$ Retraction (Split Epi) $\implies$ Epimorphism.

Corollary 2.6 (Split Morphisms are Mono/Epi)**.** *Every section (split monomorphism) is a monomorphism, and every retraction (split epimorphism) is an epimorphism. Consequently, every isomorphism is both a monomorphism and an epimorphism.*

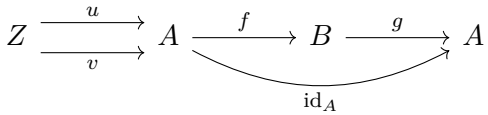


Figure 14: A section $f : A \to B$ with retraction $g : B \to A$ ($g \circ f = \mathrm{id}_A$) is automatically left-cancellable.

Diagram Explanation: In Figure 14, the identity morphism id_A is trivially a monomorphism. Since $g \circ f = \mathrm{id}_A$ is a monomorphism, Proposition 2.5(2) guarantees that the first factor f is a monomorphism.

2.3 Worked Example Problem and Practice Exercises for Section 2

Worked Example 2.7 (A Morphism that is Both Mono and Epi, Yet Not an Isomorphism [4, 3])**.** **Question:** In **Set**, every map that is both injective (mono) and surjective (epi) is a bijection (isomorphism). Is it true in *every* category that a morphism which is both a monomorphism and an epimorphism must be an isomorphism? Examine the inclusion $i : \mathbb{Z} \hookrightarrow \mathbb{Q}$ in **Ring** and the identity map $j : (\mathbb{R}, \tau_{\mathrm{disc}}) \to (\mathbb{R}, \tau_{\mathrm{std}})$ in **Top**.

Detailed Solution and Deduction: The answer is **No**! A morphism that is both mono and epi is called a *bimorphism*, and in neither **Ring** nor **Top** is every bimorphism an isomorphism:

1. **In Top:** Let $X = (\mathbb{R}, \tau_{\mathrm{disc}})$ carry the discrete topology (every subset is open) and $Y = (\mathbb{R}, \tau_{\mathrm{std}})$ carry the standard Euclidean topology. The underlying function $j(x) = x$ is continuous (since every subset of X is open), injective (hence a monomorphism in **Top**), and surjective (hence an epimorphism in **Top**). However, $j^{-1} : Y \to X$ is *not* continuous because $\{0\}$ is open in X but not open in Y. Thus j is mono and epi, but not an isomorphism in **Top**.

2. **In Ring (unital rings):** Let $i : \mathbb{Z} \hookrightarrow \mathbb{Q}$ be the inclusion. Since i is injective, it is a monomorphism. Now suppose $h_1, h_2 : \mathbb{Q} \rightrightarrows R$ are two unital ring homomorphisms such that $h_1 \circ i = h_2 \circ i$. Then for every integer $n \in \mathbb{Z}$, $h_1(n) = h_2(n)$. For any non-zero integer $m \neq 0$, $1 = m \cdot (1/m)$ in $\mathbb{Q}$, so in the ring R we have:

$$\begin{aligned} h_1(1/m) &= h_1(1/m) \cdot 1_R \\ &= h_1(1/m) \cdot h_2(m) \cdot h_2(1/m) \\ &= h_1(1/m) \cdot h_1(m) \cdot h_2(1/m) \\ &= 1_R \cdot h_2(1/m) \\ &= h_2(1/m). \end{aligned}$$

Hence

$$h_1(n/m) = h_1(n)h_1(1/m) = h_2(n)h_2(1/m) = h_2(n/m)$$

for all $n/m \in \mathbb{Q}$, proving $h_1 = h_2$. Thus $i : \mathbb{Z} \hookrightarrow \mathbb{Q}$ is an **epimorphism** in **Ring** even though it is not surjective! Clearly i is not an isomorphism since $\mathbb{Z} \ncong \mathbb{Q}$. □

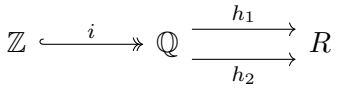


Figure 15: The ring inclusion $i : \mathbb{Z} \hookrightarrow \mathbb{Q}$ uniquely determines any ring homomorphism out of $\mathbb{Q}$.

Diagram Explanation: Figure 15 shows why epimorphisms in general categories capture "dense" or "uniquely determining" maps rather than naive set-theoretic surjectivity: once two ring homomorphisms h_1, h_2 agree on $\mathbb{Z}$, multiplicativity forces them to agree on all fractions in $\mathbb{Q}$.

Practice Exercise 2.1. Prove that in **Set**, a function $f : A \to B$ is a monomorphism if and only if it is injective, and an epimorphism if and only if it is surjective. *Hint:* For injectivity, use a singleton test object $Z = \{*\}$; for surjectivity, use a two-element test object $W = \{0, 1\}$.

Practice Exercise 2.2. Let **Haus** be the category of Hausdorff topological spaces and continuous maps. Prove that if $f : X \to Y$ is a continuous map whose image $f(X)$ is dense in Y (for example, $\mathbb{Q} \hookrightarrow \mathbb{R}$), then f is an epimorphism in **Haus**.

Conceptual Bridge to Section 3: The Obstruction Engine

Internal arrow analysis within **Top** reveals a fundamental limitation of geometry: knowing that $i : S^1 \hookrightarrow D^2$ is a monomorphism does not tell us whether a continuous retraction $r : D^2 \to S^1$ can exist. Geometry alone cannot easily prove the *non-existence* of continuous maps. To solve this, we must build a bridge from the difficult geometric world of **Top** into a computable algebraic world like **Grp** or **Ab**, where equations like $n \cdot 0 = 1$ are immediately recognized as impossible. In Section 3, we introduce the core vehicles that transport categories into one another: **Functors**.

3 Functors: Structure-Preserving Bridges Between Worlds

If categories are mathematical universes, **functors** are the passages between them [15, 2]. In Algebraic Topology, functors are the primary protagonists: they convert topological spaces and continuous maps into groups and homomorphisms so that topological questions can be solved by algebraic computation [6, 10].

Just as a group homomorphism $\phi : G \to H$ preserves the internal algebraic operation of groups ($\phi(xy) = \phi(x)\phi(y)$) and a continuous map preserves topological openness, a **functor** is a structure-preserving homomorphism between categories. It maps the algebraic skeleton of a source category $\mathcal{C}$ into a target category $\mathcal{D}$ while leaving all composition relations and identity loops intact.

Definition 3.1 (Functor as a Categorical Homomorphism [2, 3])**.** Let $\mathcal{C}$ and $\mathcal{D}$ be categories. A **functor** $F : \mathcal{C} \to \mathcal{D}$ is a rule consisting of two interrelated mappings:

1. **Object Mapping:** To each object $A \in \mathrm{Ob}(\mathcal{C})$, F assigns an object:
$$F(A) \in \mathrm{Ob}(\mathcal{D}).$$
2. **Morphism Mapping:** To each morphism $f \in \mathrm{Hom}_{\mathcal{C}}(A, B)$ between objects A and B, F assigns a morphism:
$$F(f) \in \mathrm{Hom}_{\mathcal{D}}(F(A), F(B)),$$
preserving sources and targets: $\mathrm{dom}(F(f)) = F(\mathrm{dom}(f))$ and $\mathrm{cod}(F(f)) = F(\mathrm{cod}(f))$.

These assignments are required to satisfy two foundational **functoriality axioms**:

$$F(\mathrm{id}_A) = \mathrm{id}_{F(A)} \qquad \text{for every object } A \in \mathrm{Ob}(\mathcal{C}), \tag{5}$$

$$F(g \circ f) = F(g) \circ F(f) \qquad \text{for every composable pair } A \xrightarrow{f} B \xrightarrow{g} C \text{ in } \mathcal{C}. \tag{6}$$

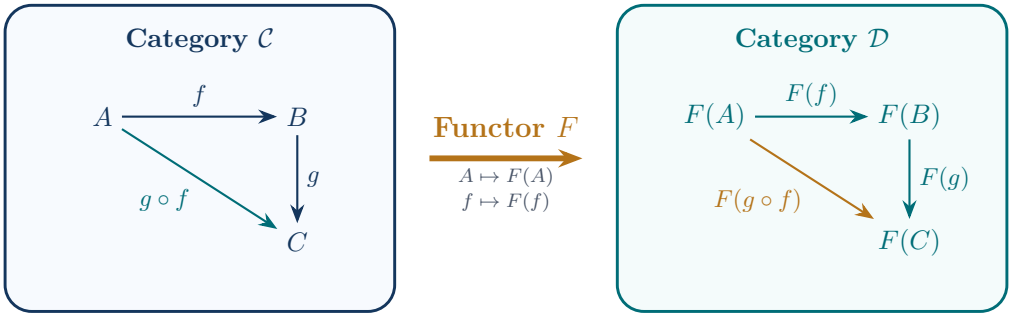


Figure 16: Macroscopic architecture of a functor $F : \mathcal{C} \to \mathcal{D}$ mapping objects, morphisms, and commutative triangles.

Diagram Explanation: In Figure 16, the functorial passage between two categories is shown:

- **Dual Action:** A functor F acts simultaneously on the objects ($A \mapsto F(A)$) and on the arrows ($f \mapsto F(f)$), ensuring that the boundary relations $\mathrm{dom}(f)$ and $\mathrm{cod}(f)$ are faithfully mapped to $\mathrm{dom}(F(f))$ and $\mathrm{cod}(F(f))$.
- **Preservation of Commutative Diagrams:** If the triangle on the left commutes in $\mathcal{C}$ (meaning $h = g \circ f$), then applying F yields:

  $$F(h) = F(g \circ f) = F(g) \circ F(f).$$

  Thus, **every commutative diagram in $\mathcal{C}$ automatically maps to a commutative diagram in $\mathcal{D}$.** This single property makes functors the foundational tools for constructing topological invariants.

In the definition above, the morphism mapping preserves the forward orientation of arrows ($f : A \to B \implies F(f) : F(A) \to F(B)$). However, many crucial algebraic invariants in topology naturally reverse arrow directions. This leads us to the fundamental distinction between covariant and contravariant functors.

The Geometric Intuition Behind Reversed Variance: Why must some algebraic invariants invert arrow directions? The answer lies in the duality between *configurations* and *measurements* (or functions on spaces). If $f : X \to Y$ is a continuous mapping:

- **Probing by Maps into the Space (Homology / Covariance):** If we map a test object (like a circle or simplex) *into* X via $\gamma : S^1 \to X$, composing with f pushes the loop forward to $f \circ \gamma : S^1 \to Y$. The loop travels in the same direction as f. Hence, fundamental groups π_1 and singular homology H_n are **covariant**.
- **Probing by Functions out of the Space (Cohomology / Contravariance):** If we assign algebraic values by evaluating real-valued functions $\psi : Y \to \mathbb{R}$ on the target, a point $x \in X$ can only be evaluated by pulling back: $(\psi \circ f)(x) = \psi(f(x))$. The function is pulled back from Y to X. Hence, differential forms, dual spaces, and singular cohomology H^n are **contravariant**.

3.1 Covariant and Contravariant Functors

When a functor preserves the forward direction of arrows as in Definition 3.1, we call it **covariant**; when it reverses arrow directions, we call it **contravariant**. To make this variance dichotomy explicit for calculations in homology and cohomology, we record both side by side:

Definition 3.2 (Covariant Functor [1, 2, 15])**.** Let $\mathcal{C}$ and $\mathcal{D}$ be categories. A **covariant functor** $F : \mathcal{C} \to \mathcal{D}$ consists of:

1. An **object assignment** $A \mapsto F(A) \in \mathrm{Ob}(\mathcal{D})$ for every object $A \in \mathrm{Ob}(\mathcal{C})$.
2. A **morphism assignment** $(f : A \to B) \mapsto (F(f) : F(A) \to F(B))$ for every morphism $f \in \mathrm{Hom}_{\mathcal{C}}(A, B)$, satisfying the two **functoriality axioms**:

$$F(g \circ f) = F(g) \circ F(f) \qquad \text{for all composable } A \xrightarrow{f} B \xrightarrow{g} C \text{ in } \mathcal{C}, \tag{7}$$

$$F(\mathrm{id}_A) = \mathrm{id}_{F(A)} \qquad \text{for every object } A \in \mathrm{Ob}(\mathcal{C}). \tag{8}$$

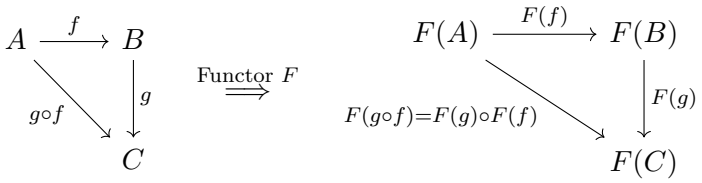


Figure 17: A covariant functor $F : \mathcal{C} \to \mathcal{D}$ preserves commutative triangles and arrow directions.

Diagram Explanation: Figure 17 shows that a covariant functor F translates a commutative triangle in the source category $\mathcal{C}$ (left) into a commutative triangle in the target category $\mathcal{D}$ (right), preserving domains, codomains, identities, and composition.

Definition 3.3 (Contravariant Functor [2, 3])**.** A **contravariant functor** F from $\mathcal{C}$ to $\mathcal{D}$ is a covariant functor $F : \mathcal{C}^{\mathrm{op}} \to \mathcal{D}$. Explicitly, F assigns to each object $A \in \mathrm{Ob}(\mathcal{C})$ an object $F(A) \in \mathrm{Ob}(\mathcal{D})$, and to each morphism $f : A \to B$ in $\mathcal{C}$ a **reversed morphism** $F(f) : F(B) \to F(A)$ in $\mathcal{D}$ such that:

$$F(g \circ f) = F(f) \circ F(g) \quad \text{and} \quad F(\mathrm{id}_A) = \mathrm{id}_{F(A)}. \tag{9}$$

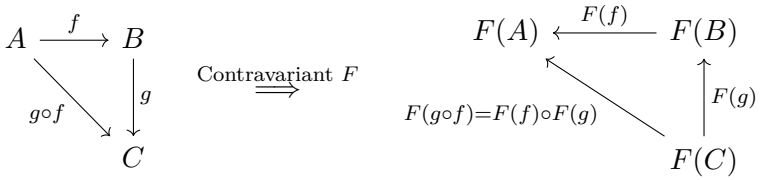


Figure 18: A contravariant functor $F : \mathcal{C}^{\mathrm{op}} \to \mathcal{D}$ reverses arrow directions and reverses composition order.

Diagram Explanation: In Figure 18, because $f : A \to B$ is mapped to $F(f) : F(B) \to F(A)$ and $g : B \to C$ is mapped to $F(g) : F(C) \to F(B)$, the composite path in $\mathcal{D}$ must start at $F(C)$, travel via $F(g)$ to $F(B)$, and then travel via $F(f)$ to $F(A)$, explaining why $F(g \circ f) = F(f) \circ F(g)$.

3.2 Key Functors in Algebra and Algebraic Topology

To understand how category theory bridges distinct mathematical disciplines, we examine four foundational families of functors that every student of algebraic topology encounters regularly.

3.2.1 Forgetful Functors: Stripping Structure

A **forgetful functor** (often denoted U, for *underlying*) is a functor that "forgets" or strips away some or all of the algebraic or topological structure of an object, retaining only the underlying set and set-theoretic functions.

Definition 3.4 (Forgetful Functors on **Grp** and **Top**)**.** A **forgetful functor** (traditionally denoted U, for *underlying*) strips away structural axioms or topological data while preserving underlying point sets and mappings:

1. **The Algebraic Forgetful Functor** ($U : \mathbf{Grp} \to \mathbf{Set}$):
   - *Object assignment*: Assigns each group $(G, \cdot, e)$ to its underlying carrier set $U(G) = G$.
   - *Morphism assignment*: Assigns each group homomorphism $f : G \to H$ to its underlying set-theoretic function $U(f) = f : G \to H$.
2. **The Topological Forgetful Functor** ($U : \mathbf{Top} \to \mathbf{Set}$):
   - *Object assignment*: Assigns each topological space (X, τ_X) to its underlying set of points $U(X) = X$, discarding the topology τ_X.
   - *Morphism assignment*: Assigns each continuous map $f : X \to Y$ to its underlying set function $U(f) = f : X \to Y$.

Because identities in **Grp** and **Top** are ordinary identity functions ($\mathrm{id}_G(x) = x$), and morphism composition is standard function composition, U strictly preserves categorical structure:

$$U(\mathrm{id}_G) = \mathrm{id}_{U(G)} \qquad \text{and} \qquad U(g \circ f) = U(g) \circ U(f).$$

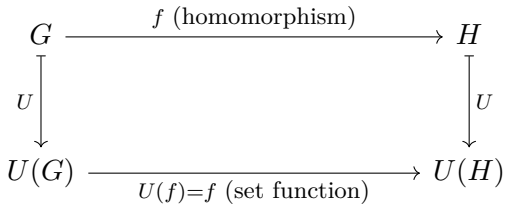


Figure 19: The forgetful functor $U : \mathbf{Grp} \to \mathbf{Set}$ projecting algebraic structure to underlying set maps.

Diagram Explanation: In Figure 19, the horizontal top arrow lives in **Grp** while the bottom arrow lives in **Set**. The vertical downward assignments (U) represent structural stripping: U retains elements and the pointwise mapping $x \mapsto f(x)$, but forgets the group compatibility law $f(xy) = f(x)f(y)$. Crucially, forgetful functors are **faithful** (they never identify distinct homomorphisms between the same pair of groups), but they are **not full** (the vast majority of set functions $G \to H$ fail to respect the group operation and thus do not lift to morphisms in **Grp**).

3.2.2 The Fundamental Group Functor $\pi_1 : \mathbf{Top}_* \to \mathbf{Grp}$

In algebraic topology, the fundamental group is the primary tool for distinguishing non-homeomorphic spaces by probing them with continuous closed loops [6, 20].

Definition 3.5 (Fundamental Group Functor)**.** The **fundamental group functor** $\pi_1 : \mathbf{Top}_* \to \mathbf{Grp}$ is defined by:

1. **On Objects:** To each pointed topological space (X, x_0), π_1 assigns the group $\pi_1(X, x_0)$ of endpoint-preserving homotopy classes $[\gamma]$ of continuous loops $\gamma : ([0,1], \{0,1\}) \to (X, x_0)$, equipped with loop concatenation $[\gamma_1] \cdot [\gamma_2] = [\gamma_1 * \gamma_2]$.
2. **On Morphisms:** To each pointed continuous map $f : (X, x_0) \to (Y, y_0)$ (satisfying $f(x_0) = y_0$), π_1 assigns the **induced group homomorphism**:

$$f_* = \pi_1(f) : \pi_1(X, x_0) \longrightarrow \pi_1(Y, y_0), \qquad f_*([\gamma]) := [f \circ \gamma].$$

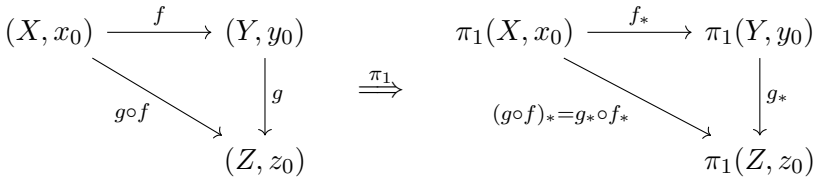


Figure 20: Functoriality of the fundamental group: a continuous topological triangle in $\mathbf{Top}_*$ maps to a commutative homomorphism triangle in $\mathbf{Grp}$.

Diagram Explanation: In Figure 20, the geometric action of π_1 is displayed:

- **Pushforward of Loops:** The continuous map f pushes a loop γ inside X forward to a loop $f \circ \gamma$ inside Y. If $\gamma \simeq \gamma'$ via a basepoint-preserving homotopy H, then $f \circ H$ is a homotopy between $f \circ \gamma$ and $f \circ \gamma'$, ensuring that f_* is well-defined.
- **Preservation of Composition:** For composable pointed maps $(X, x_0) \xrightarrow{f} (Y, y_0) \xrightarrow{g} (Z, z_0)$, the associative property of function composition gives:

$$(g \circ f)_*([\gamma]) = [(g \circ f) \circ \gamma] = [g \circ (f \circ \gamma)] = g_*\big([f \circ \gamma]\big) = (g_* \circ f_*)([\gamma]),$$

  proving that $\pi_1(g \circ f) = \pi_1(g) \circ \pi_1(f)$ without any ambiguity.

3.2.3 Singular Homology H_n and Cohomology H^n: Covariant vs. Contravariant Duality

While the fundamental group π_1 is sensitive only to 1-dimensional loops and can be non-abelian, **singular homology** and **singular cohomology** provide computable abelian invariants across all integer dimensions $n \geq 0$ [6, 7]. Crucially, they exhibit opposite categorical variances:

Definition 3.6 (Homology vs. Cohomology Functors)**.** Let $n \geq 0$ be an integer, and let G be an abelian coefficient group.

1. **Singular Homology Functor (Covariant):** $H_n : \mathbf{Top} \to \mathbf{Ab}$ assigns to each space X its n-th homology group $H_n(X) = \ker(\partial_n)/\mathrm{im}(\partial_{n+1})$. A continuous map $f : X \to Y$ acts on singular n-simplices by post-composition ($\sigma \mapsto f \circ \sigma$), inducing the **pushforward homomorphism** in the *same* direction:

$$f_* = H_n(f) : H_n(X) \longrightarrow H_n(Y), \qquad (g \circ f)_* = g_* \circ f_*.$$

2. **Singular Cohomology Functor (Contravariant):** $H^n(-; G) : \mathbf{Top}^{\mathrm{op}} \to \mathbf{Ab}$ assigns to X the cohomology group $H^n(X; G)$ of the dual cochain complex $C^n(X; G) = \mathrm{Hom}_{\mathbf{Ab}}(C_n(X), G)$. A continuous map $f : X \to Y$ acts on cochains by **pullback (pre-composition)**: a cochain $\psi : C_n(Y) \to G$ is pulled back to $\psi \circ f_\# : C_n(X) \to G$, inducing the homomorphism in the *reversed* direction:

$$f^* = H^n(f) : H^n(Y; G) \longrightarrow H^n(X; G), \qquad (g \circ f)^* = f^* \circ g^*.$$

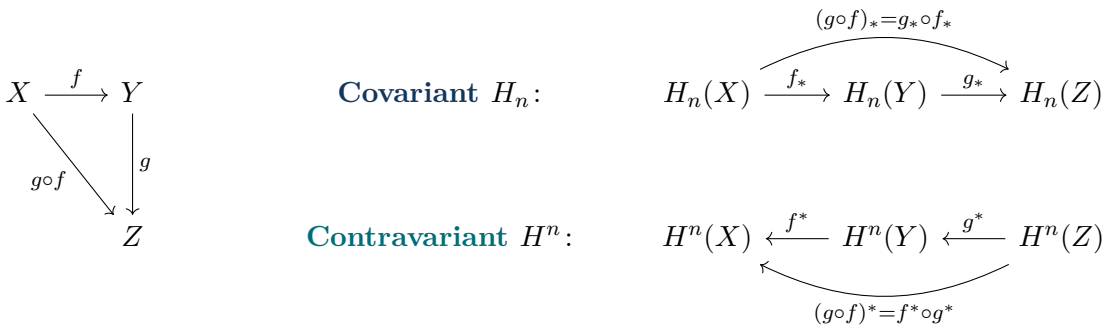


Figure 21: Topological composition in **Top** (left) induces forward composition in homology H_n (top right) and reversed composition in cohomology H^n (bottom right).

Diagram Explanation: In Figure 21, the fundamental duality between homology and cohomology is visible:

- **Homology pushes cycles forward:** f_* carries an n-cycle $z \in C_n(X)$ forward to $f_\#(z) \in C_n(Y)$, preserving the sequence of spaces $X \to Y \to Z$.
- **Cohomology pulls evaluations back:** A cohomology class on Z evaluates on chains in Z. Pulling back along g and then f requires evaluating on Y first, then on X, explaining why the order of operations inverts: $(g \circ f)^* = f^* \circ g^*$.

3.2.4 Representable Hom-Functors: The Category's Internal Radar

Every locally small category $\mathcal{C}$ possesses an intrinsic mechanism to probe its own objects using morphisms. By fixing one object $A \in \mathrm{Ob}(\mathcal{C})$, the collections $\mathrm{Hom}_{\mathcal{C}}(A, X)$ and $\mathrm{Hom}_{\mathcal{C}}(X, A)$ become functors into **Set** [4, 3].

Definition 3.7 (Covariant and Contravariant Hom-Functors)**.** Let $\mathcal{C}$ be a locally small category, and fix an object $A \in \mathrm{Ob}(\mathcal{C})$.

1. The **covariant representable functor** $h^A = \mathrm{Hom}_{\mathcal{C}}(A, -) : \mathcal{C} \to \mathbf{Set}$ is defined by:

$$X \longmapsto \mathrm{Hom}_{\mathcal{C}}(A, X), \qquad (f : X \to Y) \longmapsto \big(f_* = f \circ - : \mathrm{Hom}_{\mathcal{C}}(A, X) \to \mathrm{Hom}_{\mathcal{C}}(A, Y)\big).$$

   For any morphism $u : A \to X$, the induced map acts by **post-composition**: $f_*(u) = f \circ u$.

2. The **contravariant representable functor** $h_A = \mathrm{Hom}_{\mathcal{C}}(-, A) : \mathcal{C}^{\mathrm{op}} \to \mathbf{Set}$ is defined by:

$$X \longmapsto \mathrm{Hom}_{\mathcal{C}}(X, A), \qquad (f : X \to Y) \longmapsto \big(f^* = - \circ f : \mathrm{Hom}_{\mathcal{C}}(Y, A) \to \mathrm{Hom}_{\mathcal{C}}(X, A)\big).$$

   For any morphism $v : Y \to A$, the induced map acts by **pre-composition**: $f^*(v) = v \circ f$.

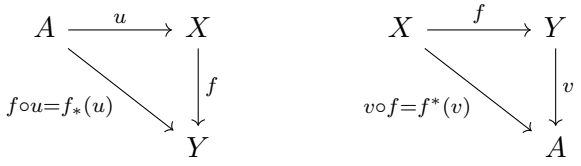


Figure 22: Left: Covariant post-composition $f_*(u) = f \circ u$ in $\mathrm{Hom}_{\mathcal{C}}(A, -)$. Right: Contravariant pre-composition $f^*(v) = v \circ f$ in $\mathrm{Hom}_{\mathcal{C}}(-, A)$.

Diagram Explanation: Figure 22 illustrates why the placement of the fixed object A dictates variance:

- **Post-composition (Left):** When A is the source, applying an arrow $f : X \to Y$ happens on the outside (post-composition), mapping $u : A \to X$ forward to $f \circ u : A \to Y$.
- **Pre-composition (Right):** When A is the target, applying $f : X \to Y$ must occur before reaching A (pre-composition), converting a morphism $v : Y \to A$ out of Y into a morphism $v \circ f : X \to A$ out of X, reversing the arrow from $X \to Y$ to a map $\mathrm{Hom}(Y, A) \to \mathrm{Hom}(X, A)$.

Connecting Hom-Functors to Topological Invariants: Notice that the fundamental group functor $\pi_1(X, x_0)$ is nothing other than a homotopy-factored representable functor! Pointed loops are continuous maps from (S^1, s_0) into (X, x_0), so the carrier set of $\pi_1(X, x_0)$ is precisely $\mathrm{Hom}_{\mathbf{hTop}_*}(S^1, X)$. Similarly, higher homotopy groups satisfy $\pi_n(X, x_0) = \mathrm{Hom}_{\mathbf{hTop}_*}(S^n, X)$. Representable functors are not abstract curiosities: they are *geometric probes* that measure an unknown space X by projecting standard geometric shapes (S^1, S^n, Δ^k) into it!

3.3 The Functorial Pipeline: Homotopy Invariance and Category Factoring

Before proving that functors preserve isomorphisms and retractions, we must address a central question that motivates all of algebraic topology: *Why are topological invariants so effective at distinguishing spaces?*

In topology, asking whether two spaces X and Y are strictly homeomorphic is notoriously difficult. Instead, we study spaces up to **homotopy equivalence** ($X \simeq Y$). The reason functors like π_1 and H_n can distinguish spaces is that they do not merely act on continuous maps; they are **homotopy-invariant functors**.

Definition 3.8 (Homotopy-Invariant Functor)**.** A functor $F : \mathbf{Top} \to \mathcal{C}$ (or $F : \mathbf{Top}_* \to \mathcal{C}$) is called **homotopy-invariant** if it assigns identical morphisms to homotopic continuous maps:

$$f \simeq g \implies F(f) = F(g) \quad \text{in } \mathrm{Hom}_{\mathcal{C}}(F(X), F(Y)).$$

Equivalently, F satisfies the universal property of the quotient category: it factors uniquely through the canonical projection functor $P : \mathbf{Top} \to \mathbf{hTop}$ (where $P(X) = X$ and $P(f) = [f]$):

$$F = \widetilde{F} \circ P, \qquad \text{for a unique functor } \widetilde{F} : \mathbf{hTop} \longrightarrow \mathcal{C}.$$

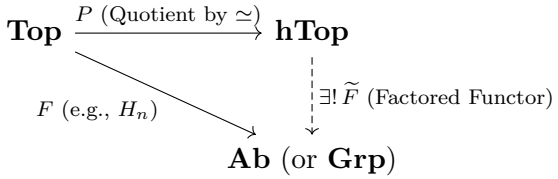


Figure 23: The Algebraic Topology Pipeline: every homotopy-invariant functor factors uniquely through the homotopy category **hTop**.

Diagram Explanation: In Figure 23, the architecture of topological invariants is formalized:

- **Geometric Simplification (P):** The projection functor $P : \mathbf{Top} \to \mathbf{hTop}$ collapses the uncountably infinite space of continuous functions down to discrete homotopy equivalence classes $[f]$.
- **Algebraic Translation ($\widetilde{F}$):** The factored functor $\widetilde{F}$ translates these homotopy classes into group homomorphisms. Because $\widetilde{F}$ is a genuine functor on **hTop**, it carries isomorphisms in **hTop** (which are precisely **homotopy equivalences** $X \simeq Y$) directly to isomorphisms in **Ab** (group isomorphisms $H_n(X) \cong H_n(Y)$).

3.3.1 Morphism Reflection and Faithful Functors

While all functors preserve compositions and identities, they do not automatically preserve or reflect all categorical properties. To navigate proofs in algebraic topology, students must distinguish three levels of fidelity:

Definition 3.9 (Preservation, Reflection, and Full/Faithful Properties)**.** Let $F : \mathcal{C} \to \mathcal{D}$ be a functor.

1. F is **faithful** if for all $A, B \in \mathrm{Ob}(\mathcal{C})$, the map on hom-sets:

$$F_{A,B} : \mathrm{Hom}_{\mathcal{C}}(A, B) \longrightarrow \mathrm{Hom}_{\mathcal{D}}(F(A), F(B)), \qquad f \longmapsto F(f)$$

   is injective (distinct arrows remain distinct).

2. F is **full** if $F_{A,B}$ is surjective (every arrow between $F(A)$ and $F(B)$ in $\mathcal{D}$ comes from an arrow in $\mathcal{C}$).

3. F **reflects isomorphisms** if whenever $F(f)$ is an isomorphism in $\mathcal{D}$, the original arrow f is guaranteed to be an isomorphism in $\mathcal{C}$.

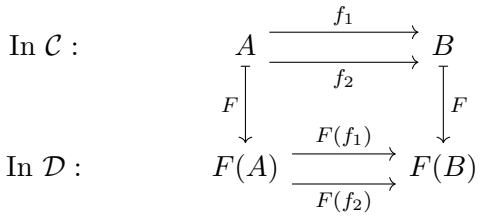


Figure 24: Faithful functors keep parallel arrows distinct: $f_1 \neq f_2 \implies F(f_1) \neq F(f_2)$.

Diagram Explanation: In Figure 24, the distinction between preservation and reflection is illustrated:

- Functors always **preserve** equations: if $f_1 = f_2$, then $F(f_1) = F(f_2)$.
- But functors do not generally **reflect** equations: if $F(f_1) = F(f_2)$, we cannot deduce $f_1 = f_2$ unless F is faithful.
- In topological data analysis and homology, H_n is *not* faithful (many non-homotopic maps induce the zero homomorphism), which is precisely why it simplifies geometric complexity into manageable algebra!

With these foundational concepts established, we now prove the central algebraic result that powers all topological obstruction theorems: functors unconditionally preserve commutative equations, isomorphisms, and retraction structures.

Proposition 3.10 (Functors Preserve Isomorphisms, Retractions, and Sections)**.** *Let $F : \mathcal{C} \to \mathcal{D}$ be any functor (covariant or contravariant).*

1. *If $f : A \to B$ is an isomorphism in $\mathcal{C}$, then $F(f)$ is an isomorphism in $\mathcal{D}$ with $F(f)^{-1} = F(f^{-1})$.*
2. *If $r : X \to A$ is a retraction in $\mathcal{C}$ with section $i : A \to X$ ($r \circ i = \mathrm{id}_A$), and F is covariant, then $F(r) : F(X) \to F(A)$ is a retraction with section $F(i) : F(A) \to F(X)$.*

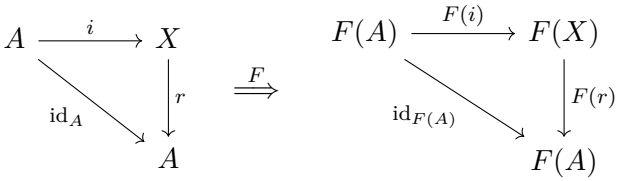


Figure 25: A functor F maps a retraction diagram $r \circ i = \mathrm{id}_A$ in $\mathcal{C}$ to a retraction diagram in $\mathcal{D}$.

Diagram Explanation: In Figure 25, because $F(r) \circ F(i) = F(r \circ i) = F(\mathrm{id}_A) = \mathrm{id}_{F(A)}$, the homomorphism $F(i)$ must be injective (a section) and $F(r)$ must be surjective (a retraction). If no such injective homomorphism $F(A) \hookrightarrow F(X)$ can exist in $\mathcal{D}$, then no continuous retraction $r : X \to A$ could have existed in $\mathcal{C}$!

Proof. (1) If $g = f^{-1}$ satisfies $g \circ f = \mathrm{id}_A$ and $f \circ g = \mathrm{id}_B$, then for covariant F, Equations (7)–(8) give $F(g) \circ F(f) = F(g \circ f) = F(\mathrm{id}_A) = \mathrm{id}_{F(A)}$ and $F(f) \circ F(g) = \mathrm{id}_{F(B)}$. (For contravariant F, the roles of left and right inverse simply swap.) (2) Similarly, $r \circ i = \mathrm{id}_A \implies F(r) \circ F(i) = F(r \circ i) = F(\mathrm{id}_A) = \mathrm{id}_{F(A)}$. □

The Logic of Obstruction Theory (Why Topologists Use the Contrapositive): Because functors like π_1 and H_n are *not* full or faithful (Definition 3.9), they do not reflect isomorphisms: having $\pi_1(X) \cong \pi_1(Y)$ does *not* guarantee $X \simeq Y$. However, because functors *preserve* isomorphisms (Proposition 3.10), their contrapositive provides an ironclad **obstruction test** for proving two spaces are topologically distinct:

Corollary 3.11 (Topological Invariance via Functors [6])**.** *If two pointed spaces (X, x_0) and (Y, y_0) have non-isomorphic fundamental groups $\pi_1(X, x_0) \ncong \pi_1(Y, y_0)$ (or non-isomorphic homology groups $H_n(X) \ncong H_n(Y)$ for some n), then X and Y are neither homeomorphic nor homotopy equivalent.*

$$(X, x_0) \xrightarrow{\text{no } \simeq} (Y, y_0) \qquad \overset{\text{contrapositive}}{\Longleftarrow} \qquad \pi_1(X, x_0) \xrightarrow{\text{no } \cong} \pi_1(Y, y_0)$$

Figure 26: Contrapositive of Proposition 3.10: distinct algebraic invariants rule out topological equivalence.

Diagram Explanation: Figure 26 summarizes the core strategy of Algebraic Topology: because π_1 and H_n factor as functors through $\mathbf{hTop}_*$ and $\mathbf{hTop}$, Proposition 3.10(1) implies that any homotopy equivalence $X \simeq Y$ forces $\pi_1(X, x_0) \cong \pi_1(Y, y_0)$ and $H_n(X) \cong H_n(Y)$.

3.4 Worked Example Problem and Practice Exercises for Section 3

Worked Example 3.12 (Brouwer's No-Retraction Theorem and Fixed-Point Theorem via Functoriality [6, 4])**. Question:** Using the fact that $\pi_1(S^1, x_0) \cong \mathbb{Z}$ and $\pi_1(D^2, x_0) \cong \{0\}$ (where $D^2 = \{x \in \mathbb{R}^2 : \|x\| \leq 1\}$ is the closed unit disk and $S^1 = \partial D^2$ is its boundary circle), apply Proposition 3.10(2) to prove: (a) There is no continuous retraction $r : D^2 \to S^1$. (b) Every continuous map $f : D^2 \to D^2$ has at least one fixed point $x^* \in D^2$ such that $f(x^*) = x^*$ (the 2D Brouwer Fixed-Point Theorem).

Detailed Solution and Deduction:

1. **Part (a) — No Retraction $D^2 \to S^1$:** Suppose for the sake of contradiction that there exists a continuous retraction $r : D^2 \to S^1$, meaning $r \circ i = \mathrm{id}_{S^1}$ where $i : S^1 \hookrightarrow D^2$ is the boundary inclusion. Choosing a basepoint $x_0 \in S^1$, both i and r are morphisms in $\mathbf{Top}_*$. Applying the covariant functor $\pi_1 : \mathbf{Top}_* \to \mathbf{Grp}$ (Proposition 3.10(2)), we obtain a commutative triangle in $\mathbf{Grp}$:
$$\pi_1(r) \circ \pi_1(i) = \pi_1(r \circ i) = \pi_1(\mathrm{id}_{S^1}) = \mathrm{id}_{\pi_1(S^1, x_0)}.$$
However, substituting $\pi_1(S^1, x_0) \cong \mathbb{Z}$ and $\pi_1(D^2, x_0) \cong \{0\}$, the homomorphism $\pi_1(i) : \mathbb{Z} \to \{0\}$ must send every integer to 0, and $\pi_1(r) : \{0\} \to \mathbb{Z}$ must send 0 to 0. Thus the composite $\pi_1(r) \circ \pi_1(i) : \mathbb{Z} \to \mathbb{Z}$ is the zero map $n \mapsto 0$, which cannot equal the identity map $\mathrm{id}_{\mathbb{Z}}(1) = 1$! This contradiction proves that no continuous retraction $r : D^2 \to S^1$ exists.

2. **Part (b) — Brouwer Fixed-Point Theorem:** Suppose $f : D^2 \to D^2$ is continuous and has *no* fixed point ($f(x) \neq x$ for all $x \in D^2$). For each $x \in D^2$, construct the unique ray starting at $f(x)$ and passing through x, and let $r(x) \in S^1$ be the point where this ray intersects the boundary circle S^1. Because f is continuous and $f(x) \neq x$ everywhere, the intersection formula is a continuous function of x, and for every $x \in S^1$ the ray immediately hits x itself, so $r(x) = x$. Thus $r : D^2 \to S^1$ is a continuous retraction, contradicting Part (a)! Therefore, f must have a fixed point. □

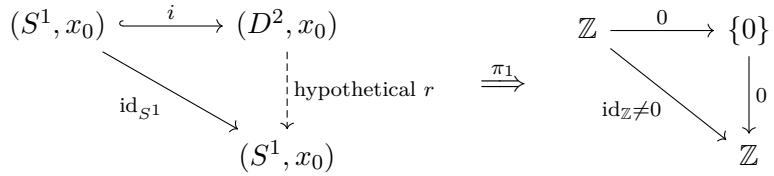


Figure 27: Functorial proof of Brouwer's No-Retraction Theorem via $\pi_1 : \mathbf{Top}_* \to \mathbf{Grp}$.

Diagram Explanation: Figure 27 shows the power of Proposition 3.10: the impossible algebraic factorization of $\mathrm{id}_{\mathbb{Z}} : \mathbb{Z} \to \mathbb{Z}$ through the trivial group $\{0\}$ on the right immediately rules out the existence of the continuous geometric retraction $r : D^2 \to S^1$ on the left.

Practice Exercise 3.1 (Full and Faithful Functors [2, 4])**.** A functor $F : \mathcal{C} \to \mathcal{D}$ is called **faithful** if for every pair $A, B \in \mathrm{Ob}(\mathcal{C})$, the induced map on hom-sets $F_{A,B} : \mathrm{Hom}_{\mathcal{C}}(A, B) \to \mathrm{Hom}_{\mathcal{D}}(F(A), F(B))$ is injective, and **full** if $F_{A,B}$ is surjective for all A, B. (a) Prove that the forgetful functor $U : \mathbf{Grp} \to \mathbf{Set}$ is faithful, but not full. (b) Prove that if $F : \mathcal{C} \to \mathcal{D}$ is both full and faithful ("fully faithful"), then a morphism $f : A \to B$ is an isomorphism in $\mathcal{C}$ *if and only if* $F(f)$ is an isomorphism in $\mathcal{D}$ (i.e., F reflects isomorphisms).

Practice Exercise 3.2 (Abelianization Functor)**.** For any group G, let $[G, G]$ be its commutator subgroup (generated by all elements $xyx^{-1}y^{-1}$) and define $G^{\mathrm{ab}} = G/[G, G]$. Show that the assignment $G \mapsto G^{\mathrm{ab}}$ extends to a covariant functor $(-)^{\mathrm{ab}} : \mathbf{Grp} \to \mathbf{Ab}$.

Conceptual Bridge to Section 4: Comparing Invariants

Functors assign algebraic invariants to topological spaces. But in algebraic topology, we often compute *two different invariants for the same space*—for instance, the fundamental group $\pi_1(X, x_0)$ and the first homology group $H_1(X)$. How do these two invariants relate? Can we map $\pi_1(X)$ to $H_1(X)$ in a way that respects continuous maps $f : X \to Y$? If we choose a different basis or coordinate system, does our comparison fall apart? To compare entire functors across all spaces simultaneously without making arbitrary choices, we require morphisms between functors: **Natural Transformations**.

4 Natural Transformations and Equivalence of Categories

As Eilenberg and Mac Lane famously wrote [1, 2], they did not invent categories merely to study categories; they invented categories in order to define functors, and they invented functors in order to define **natural transformations** [16, 3].

4.1 Natural Transformations and Natural Isomorphisms

Definition 4.1 (Natural Transformation and Natural Isomorphism [1, 16])**.** Let $F, G : \mathcal{C} \to \mathcal{D}$ be two parallel covariant functors between the same categories $\mathcal{C}$ and $\mathcal{D}$. A **natural transformation** $\eta : F \Rightarrow G$ consists of a family of morphisms in $\mathcal{D}$ indexed by the objects of $\mathcal{C}$,

$$\eta = \{\eta_X : F(X) \longrightarrow G(X)\}_{X \in \mathrm{Ob}(\mathcal{C})},$$

called the **components** of η, such that for every morphism $f : X \to Y$ in $\mathcal{C}$, the following **naturality square** commutes in $\mathcal{D}$:

$$\eta_Y \circ F(f) = G(f) \circ \eta_X. \tag{10}$$

If every component $\eta_X : F(X) \to G(X)$ is an isomorphism in $\mathcal{D}$, then η is called a **natural isomorphism** (or **natural equivalence**), and we write $F \cong G$.

$$\begin{array}{ccc} X & & \begin{array}{ccc} F(X) & \xrightarrow{\eta_X} & G(X) \\ {\scriptstyle F(f)}\downarrow & & \downarrow{\scriptstyle G(f)} \\ F(Y) & \xrightarrow[\eta_Y]{} & G(Y) \end{array} \\ {\scriptstyle f}\downarrow & & \\ Y & & \end{array}$$

Figure 28: The naturality square (right) in $\mathcal{D}$ induced by a morphism $f : X \to Y$ (left) in $\mathcal{C}$.

Diagram Explanation: In Figure 28, the natural transformation $\eta : F \Rightarrow G$ translates each object $F(X)$ to $G(X)$ via the horizontal component arrow η_X. Equation (10) demands that translating from F to G at X and then applying $G(f)$ (top-then-right path) yields the exact same result as first applying $F(f)$ and then translating from F to G at Y (left-then-bottom path). Intuitively, η_X is constructed by a uniform, "choice-free" rule that works identically across all objects X.

Climbing the Categorical Ladder (Functors as Objects, Natural Transformations as Arrows): Having defined categories (0-level), functors between categories (1-level), and natural transformations between functors (2-level), a remarkable synthesis occurs: if we fix a small source category $\mathcal{C}$ and a target category $\mathcal{D}$, can we treat the functors $F, G, H : \mathcal{C} \to \mathcal{D}$ as *objects* in their own right, and natural transformations $\eta : F \Rightarrow G$ as *morphisms* between them? To do so, we must define how two natural transformations compose:

Proposition 4.2 (Vertical Composition and Functor Categories [2, 4])**.** *Let $F, G, H : \mathcal{C} \to \mathcal{D}$ be covariant functors, and let $\eta : F \Rightarrow G$ and $\theta : G \Rightarrow H$ be natural transformations.*

1. *Defining componentwise composition $(\theta \circ \eta)_X := \theta_X \circ \eta_X : F(X) \to H(X)$ yields a natural transformation $\theta \circ \eta : F \Rightarrow H$.*
2. *For any small category $\mathcal{C}$ and any category $\mathcal{D}$, the collection of functors $\mathcal{C} \to \mathcal{D}$ as objects and natural transformations as morphisms forms a category, called the* ***functor category*** *$[\mathcal{C}, \mathcal{D}]$ (or $\mathbf{Fun}(\mathcal{C}, \mathcal{D})$).*
3. *A natural transformation $\eta : F \Rightarrow G$ is an isomorphism in $[\mathcal{C}, \mathcal{D}]$ if and only if each component $\eta_X : F(X) \to G(X)$ is an isomorphism in $\mathcal{D}$.*

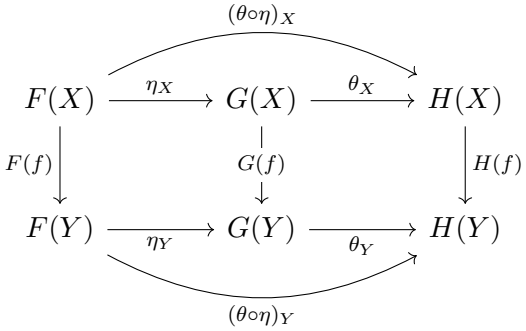


Figure 29: Pasting two adjacent naturality squares proves naturality of vertical composition $\theta \circ \eta$.

Diagram Explanation: Figure 29 demonstrates one of the most fundamental diagram-chasing techniques: **pasting commutative squares**. Because both the left square (η) and the right square (θ) commute, the outer rectangle automatically commutes: $H(f) \circ \theta_X \circ \eta_X = \theta_Y \circ G(f) \circ \eta_X = \theta_Y \circ \eta_Y \circ F(f)$.

Proof. (1) For any $f : X \to Y$ in $\mathcal{C}$, using Axiom 1.2 and the naturality of θ and η (Equation (10)):

$$H(f)\circ(\theta\circ\eta)_X = (H(f)\circ\theta_X)\circ\eta_X = (\theta_Y\circ G(f))\circ\eta_X = \theta_Y\circ(G(f)\circ\eta_X) = \theta_Y\circ(\eta_Y\circ F(f)) = (\theta\circ\eta)_Y\circ F(f).$$

(2) Associativity and identity laws in $[\mathcal{C}, \mathcal{D}]$ follow immediately from those in $\mathcal{D}$ applied at each component X, with $(\mathrm{id}_F)_X = \mathrm{id}_{F(X)}$. (3) If each η_X is an isomorphism with inverse $\eta_X^{-1} : G(X) \to F(X)$, applying η_Y^{-1} on the left and η_X^{-1} on the right of $\eta_Y \circ F(f) = G(f) \circ \eta_X$ yields $F(f) \circ \eta_X^{-1} = \eta_Y^{-1} \circ G(f)$, proving that $\{\eta_X^{-1}\}_X$ is a natural transformation $\eta^{-1} : G \Rightarrow F$. □

Why Isomorphism of Categories Violates Mathematical Practice: In geometry, we consider two spaces identical if they are homeomorphic, even if their underlying set of coordinates differ. In category theory, demanding that two categories be strictly isomorphic ($G(F(X)) = X$ on the nose)

is an unnatural demand: it requires identical object labeling. For example, consider the category of all finite sets **FinSet** versus the subcategory $\mathcal{S}$ whose objects are strictly the initial segments $\mathbf{n} = \{1, 2, \ldots, n\}$. Every finite set is isomorphic to some $\mathbf{n}$, so $\mathcal{S}$ captures all combinatorial properties of **FinSet**. Yet $\mathcal{S}$ has countably many objects, while **FinSet** is a proper class! They cannot be isomorphic. What matters is not that $G(F(X))$ equals X, but that $G(F(X))$ is *naturally isomorphic* to X. This relaxed, geometrically authentic notion is **equivalence of categories**.

4.2 Equivalence of Categories

In practice, requiring two categories $\mathcal{C}$ and $\mathcal{D}$ to be *isomorphic* (meaning $G \circ F = \mathrm{id}_{\mathcal{C}}$ and $F \circ G = \mathrm{id}_{\mathcal{D}}$ on the nose) is far too rigid because objects in categories are usually only determined up to isomorphism [4, 5].

Definition 4.3 (Equivalence of Categories [2, 3])**.** Two categories $\mathcal{C}$ and $\mathcal{D}$ are **equivalent** (written $\mathcal{C} \simeq \mathcal{D}$) if there exist functors $F : \mathcal{C} \to \mathcal{D}$ and $G : \mathcal{D} \to \mathcal{C}$ together with natural isomorphisms:

$$\eta : \mathrm{id}_{\mathcal{C}} \xRightarrow{\sim} G \circ F \quad \text{and} \quad \varepsilon : F \circ G \xRightarrow{\sim} \mathrm{id}_{\mathcal{D}}.$$

$$G\circ F \cong \mathrm{id}_{\mathcal{C}} \circlearrowright \mathcal{C} \underset{G}{\overset{F}{\rightleftarrows}} \mathcal{D} \circlearrowright F\circ G \cong \mathrm{id}_{\mathcal{D}}$$

Figure 30: An equivalence of categories $\mathcal{C} \simeq \mathcal{D}$ replaces strict equality of round-trips with natural isomorphism.

Diagram Explanation: In Figure 30, traveling from $\mathcal{C}$ to $\mathcal{D}$ via F and returning via G does not need to land on the exact same object $X \in \mathrm{Ob}(\mathcal{C})$; instead, $G(F(X))$ is naturally isomorphic to X via $\eta_X : X \xrightarrow{\sim} G(F(X))$. A famous theorem states that $F : \mathcal{C} \to \mathcal{D}$ is an equivalence of categories if and only if F is **fully faithful** (Exercise 3.1) and **essentially surjective** (every $D \in \mathrm{Ob}(\mathcal{D})$ is isomorphic to $F(C)$ for some $C \in \mathrm{Ob}(\mathcal{C})$) [2].

4.3 Worked Example Problems and Practice Exercises for Section 4

Worked Example 4.4 (The Hurewicz Homomorphism as a Natural Transformation [6, 7])**. Question:** For every pointed topological space (X, x_0), let $\pi_1(X, x_0)$ be its fundamental group and $H_1(X)$ be its first singular homology group (viewed in **Grp** via the inclusion $\mathbf{Ab} \hookrightarrow \mathbf{Grp}$). Every loop $\gamma : [0, 1] \to X$ at x_0 is a singular 1-simplex whose boundary is $\partial\gamma = \gamma(1) - \gamma(0) = x_0 - x_0 = 0$, so γ is a 1-cycle representing a homology class $[\gamma]_{H_1} \in H_1(X)$. Prove that the family of **Hurewicz maps** $h_{(X,x_0)} : \pi_1(X, x_0) \to H_1(X)$ given by $[\gamma]_{\pi_1} \mapsto [\gamma]_{H_1}$ defines a natural transformation $h : \pi_1 \Rightarrow H_1 \circ U$, where $U : \mathbf{Top}_* \to \mathbf{Top}$ forgets the basepoint.

Detailed Solution and Deduction: Let $f : (X, x_0) \to (Y, y_0)$ be any pointed continuous map in $\mathbf{Top}_*$. We must check that the naturality square (Equation (10)) commutes for every element $[\gamma]_{\pi_1} \in \pi_1(X, x_0)$:

1. **Top-then-Right Path ($H_1(f) \circ h_{(X,x_0)}$):** First applying $h_{(X,x_0)}$ sends the loop class $[\gamma]_{\pi_1}$ to the singular 1-homology class $[\gamma]_{H_1} \in H_1(X)$. Next, by definition of the induced map on singular homology, $H_1(f)$ acts on a singular 1-simplex $\gamma : \Delta^1 \to X$ by post-composition with f, sending $[\gamma]_{H_1} \mapsto [f \circ \gamma]_{H_1} \in H_1(Y)$.

2. **Left-then-Bottom Path ($h_{(Y,y_0)} \circ \pi_1(f)$):** First applying $\pi_1(f)$ sends the loop class $[\gamma]_{\pi_1} \in \pi_1(X, x_0)$ to $[f \circ \gamma]_{\pi_1} \in \pi_1(Y, y_0)$. Next, applying $h_{(Y,y_0)}$ sends $[f \circ \gamma]_{\pi_1}$ to the singular 1-homology class $[f \circ \gamma]_{H_1} \in H_1(Y)$.

Since both paths yield $[f \circ \gamma]_{H_1}$ for every $[\gamma]_{\pi_1} \in \pi_1(X, x_0)$, we have $H_1(f) \circ h_{(X,x_0)} = h_{(Y,y_0)} \circ \pi_1(f)$. Hence the Hurewicz homomorphism is a natural transformation! (By Hurewicz's Theorem [6], when X is path-connected, $h_{(X,x_0)}$ factors as an isomorphism from the abelianization $\pi_1(X, x_0)^{\mathrm{ab}} \xrightarrow{\sim} H_1(X)$.) □

$$\begin{array}{ccc} \pi_1(X, x_0) & \xrightarrow{h_{(X,x_0)}} & H_1(X) \\ {\scriptstyle \pi_1(f)}\downarrow & & \downarrow{\scriptstyle H_1(f)} \\ \pi_1(Y, y_0) & \xrightarrow[h_{(Y,y_0)}]{} & H_1(Y) \end{array}$$

Figure 31: Naturality square for the Hurewicz homomorphism $h : \pi_1 \Rightarrow H_1 \circ U$.

Diagram Explanation: Figure 31 shows that converting a homotopy class of loops into a homology class of 1-cycles commutes with pushing loops forward along any continuous map $f : (X, x_0) \to (Y, y_0)$.

Worked Example 4.5 (Double Dual of a Vector Space: Natural vs. Unnatural Isomorphism [1, 3])**.** **Question:** Let $\mathbf{Vect}_k$ be the category of vector spaces over a field k. For each $V \in \mathrm{Ob}(\mathbf{Vect}_k)$, let $V^* = \mathrm{Hom}_k(V, k)$ be its dual space and $V^{**} = (V^*)^*$ be its double dual. Define $\mathrm{ev}_V : V \to V^{**}$ by $\mathrm{ev}_V(v)(\varphi) = \varphi(v)$ for $v \in V$ and $\varphi \in V^*$. Prove that $\mathrm{ev} : \mathrm{id}_{\mathbf{Vect}_k} \Rightarrow (-)^{**}$ is a natural transformation (and a natural isomorphism on finite-dimensional vector spaces), and explain why $V \cong V^*$ for finite-dimensional V is *not* natural.

Detailed Solution and Deduction: Let $T : V \to W$ be any linear map in $\mathbf{Vect}_k$. Recall that $T^* : W^* \to V^*$ is given by $T^*(\psi) = \psi \circ T$, and $T^{**} : V^{**} \to W^{**}$ is given by $T^{**}(\Phi)(\psi) = \Phi(T^*(\psi)) = \Phi(\psi \circ T)$. For any $v \in V$ and $\psi \in W^*$, we compute:

$$\big(T^{**}(\mathrm{ev}_V(v))\big)(\psi) = \mathrm{ev}_V(v)(\psi \circ T) = (\psi \circ T)(v) = \psi(T(v)) = \big(\mathrm{ev}_W(T(v))\big)(\psi).$$

Thus $T^{**} \circ \mathrm{ev}_V = \mathrm{ev}_W \circ T$, so ev is a natural transformation! On the subclass $\mathbf{FDVect}_k$ of finite-dimensional vector spaces, $\dim V = \dim V^{**}$ and ev_V is injective, so ev_V is a linear isomorphism for every V, making ev a natural isomorphism without choosing any basis. By contrast, $(-)^* : \mathbf{FDVect}_k^{\mathrm{op}} \to \mathbf{FDVect}_k$ is *contravariant* while id is covariant, and constructing an isomorphism $V \xrightarrow{\sim} V^*$ requires choosing an arbitrary basis for V. □

$$\begin{array}{ccc} V & \xrightarrow{\mathrm{ev}_V} & V^{**} \\ {\scriptstyle T}\downarrow & & \downarrow{\scriptstyle T^{**}} \\ W & \xrightarrow[\mathrm{ev}_W]{} & W^{**} \end{array}$$

Figure 32: Naturality of the canonical evaluation map $\mathrm{ev}_V : V \to V^{**}$ in $\mathbf{Vect}_k$.

Diagram Explanation: Figure 32 is the historical example that inspired Eilenberg and Mac Lane [1]: ev_V requires no choice of basis, so it commutes with every linear transformation $T : V \to W$.

Practice Exercise 4.1 (Determinant as a Natural Transformation [4])**.** Fix an integer $n \geq 1$. Consider the functor $\mathrm{GL}_n : \mathbf{CRing} \to \mathbf{Grp}$ sending each commutative ring R to the group $\mathrm{GL}_n(R)$ of invertible $n \times n$ matrices over R, and the functor $(-)^\times : \mathbf{CRing} \to \mathbf{Grp}$ sending R to its group of units $R^\times$. Prove that the determinant $\det_R : \mathrm{GL}_n(R) \to R^\times$ defines a natural transformation $\det : \mathrm{GL}_n \Rightarrow (-)^\times$.

Practice Exercise 4.2 (Equivalence of Skeletons)**.** Let **FinSet** be the category of all finite sets and functions, and let $\mathcal{S}$ be the full subcategory whose objects are solely the standard sets $\mathbf{n} = \{1, 2, \ldots, n\}$ for $n \in \mathbb{N}_0$. Prove that the inclusion functor $I : \mathcal{S} \hookrightarrow \mathbf{FinSet}$ is an equivalence of categories (Definition 4.3), even though $\mathcal{S}$ has only countably many objects while **FinSet** is a proper class.

Conceptual Bridge to Section 5: Assembling Spaces Without Elements

In point-set topology, spaces are built by assembling sets of points: Cartesian products $X \times Y = \{(x, y)\}$, disjoint unions $X \sqcup Y$, and quotient spaces $X/\sim$. But in modern geometry, spaces are frequently presented through their mapping behaviors—for example, attaching an n-dimensional cell along its boundary $\partial D^n \to X$, or gluing overlapping open sets $U \cup V$. In Section 5, we discover that all these geometric constructions are special cases of a single arrow-theoretic principle: **Universal Mapping Properties**. We will characterize objects not by what they contain, but by the unique ways all other objects must map into them (**limits**) or out of them (**colimits**).

5 Universal Properties, Limits, and Colimits

How do we define the direct product of groups $G \times H$, the product topology on $X \times Y$, the disjoint union $X \amalg Y$, the wedge sum $X \vee Y$, or the quotient space X/A using only arrows? The answer lies in **universal properties**: we characterize an object by how all other objects map *into* it (limits) or *out of* it (colimits) [18, 4, 3].

5.1 Initial and Terminal Objects

Definition 5.1 (Initial, Terminal, and Zero Objects [2, 5])**.** Let $\mathcal{C}$ be a category.

1. An object $I \in \mathrm{Ob}(\mathcal{C})$ is **initial** if for every object $X \in \mathrm{Ob}(\mathcal{C})$, there exists a *unique* morphism $!_X : I \to X$.
2. An object $T \in \mathrm{Ob}(\mathcal{C})$ is **terminal** (or **final**) if for every object $X \in \mathrm{Ob}(\mathcal{C})$, there exists a *unique* morphism $!^X : X \to T$.
3. If an object $0 \in \mathrm{Ob}(\mathcal{C})$ is simultaneously initial and terminal, it is called a **zero object**.

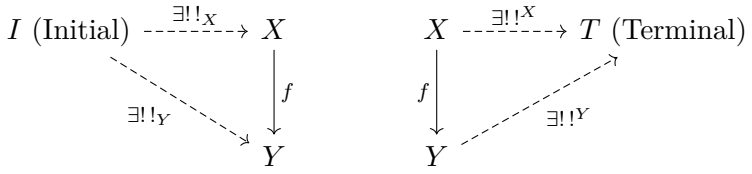


Figure 33: Universal mapping properties of an initial object I (left) and a terminal object T (right).

Diagram Explanation: In Figure 33, dashed arrows labelled ∃! denote morphisms whose existence and uniqueness are guaranteed by a universal property. Because there is only one arrow $I \to Y$, any composite $f \circ !_X : I \to Y$ must equal $!_Y$. In **Set** and **Top**, the empty set $\varnothing$ is initial and any one-point space $\{*\}$ is terminal; in **Grp**, **Ab**, and $\mathbf{Top}_*$, the one-element trivial group $\{e\}$ (or one-point space $\{*\}$) is a **zero object**!

Proposition 5.2 (Uniqueness of Initial and Terminal Objects up to Unique Isomorphism)**.** *If I and I' are both initial objects in a category $\mathcal{C}$, then there exists a* unique *isomorphism $u : I \xrightarrow{\sim} I'$. Dually, any two terminal objects are uniquely isomorphic.*

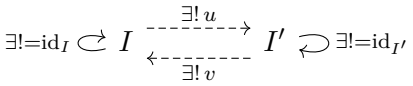


Figure 34: Unique isomorphism proof between two initial objects I and I'.

Diagram Explanation: Figure 34 illustrates the most important proof paradigm in category theory: because I is initial, there is a unique arrow $u : I \to I'$ and a unique arrow $I \to I$ (which must be id_I). Because I' is initial, there is a unique arrow $v : I' \to I$ and a unique arrow $I' \to I'$ (which must be $\mathrm{id}_{I'}$). Therefore, the round-trip $v \circ u : I \to I$ has no choice but to equal id_I, and $u \circ v = \mathrm{id}_{I'}$!

Proof. Since I is initial, there is a unique morphism $u : I \to I'$. Since I' is initial, there is a unique morphism $v : I' \to I$. Now consider the composite $v \circ u : I \to I$. Since I is initial, $\mathrm{Hom}_{\mathcal{C}}(I, I)$ contains only one morphism; since $\mathrm{id}_I \in \mathrm{Hom}_{\mathcal{C}}(I, I)$, we must have $v \circ u = \mathrm{id}_I$. By exact symmetry, $u \circ v = \mathrm{id}_{I'}$. Thus u is the unique isomorphism $I \xrightarrow{\sim} I'$. The statement for terminal objects follows by duality in $\mathcal{C}^{\mathrm{op}}$. □

5.2 Products and Coproducts

How are products and coproducts related to terminal and initial objects? A categorical product $A \times B$ is nothing other than a **terminal object in the category of spans** $A \leftarrow Z \rightarrow B$, while a coproduct $A \amalg B$ is an **initial object in the category of cospans** $A \rightarrow W \leftarrow B$! Consequently, Proposition 5.2 automatically guarantees that products and coproducts—whenever they exist—are unique up to a unique isomorphism.

Definition 5.3 (Categorical Product and Coproduct [2, 18])**.** Let A and B be objects in a category $\mathcal{C}$.

1. A **product** of A and B is an object $A \times B$ equipped with two projection morphisms $\pi_1 : A \times B \to A$ and $\pi_2 : A \times B \to B$ such that for every object $Z \in \mathrm{Ob}(\mathcal{C})$ and every pair of morphisms $f : Z \to A$ and $g : Z \to B$, there exists a *unique* morphism $\langle f, g \rangle : Z \to A \times B$ making the left diagram in Figure 35 commute: $\pi_1 \circ \langle f, g \rangle = f$ and $\pi_2 \circ \langle f, g \rangle = g$.

2. Dually, a **coproduct** (or **categorical sum**) of A and B is an object $A \amalg B$ equipped with two inclusion (injection) morphisms $\iota_1 : A \to A \amalg B$ and $\iota_2 : B \to A \amalg B$ such that for every object $W \in \mathrm{Ob}(\mathcal{C})$ and every pair of morphisms $u : A \to W$ and $v : B \to W$, there exists a *unique* morphism $[u, v] : A \amalg B \to W$ making the right diagram in Figure 35 commute: $[u, v] \circ \iota_1 = u$ and $[u, v] \circ \iota_2 = v$.

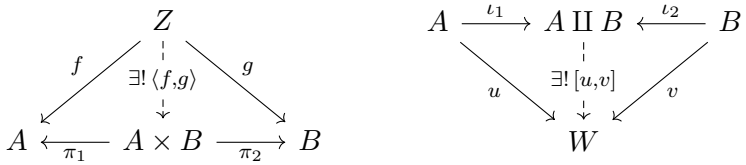


Figure 35: Universal properties of the Categorical Product $A \times B$ (left) and Coproduct $A \amalg B$ (right).

Diagram Explanation: In Figure 35 (left), mapping *into* the product $A \times B$ from a test object Z is equivalent to giving a pair of maps $(f : Z \to A, g : Z \to B)$. In Figure 35 (right), all arrows are reversed: mapping *out of* the coproduct $A \amalg B$ into a test object W is equivalent to giving a pair of maps $(u : A \to W, v : B \to W)$. Table 3 compares products and coproducts across major categories.

From Disjoint Unions to Geometric Gluing: The coproduct $A \amalg B$ joins two spaces in complete isolation, leaving them disjoint. But topology is the science of continuous connectivity: we

Category $\mathcal{C}$	Categorical Product $A \times B$	Categorical Coproduct $A \amalg B$
Set	Cartesian product $A \times B$	Disjoint union $A \sqcup B$
Top	Product space $X \times Y$ (product topology)	Disjoint union space $X \sqcup Y$
$\mathbf{Top}_*$	Pointed product $(X \times Y, (x_0, y_0))$	**Wedge sum** $X \vee Y = (X \sqcup Y)/(x_0 \sim y_0)$
Grp	Direct product of groups $G \times H$	**Free product** of groups $G * H$
Ab or R-**Mod**	Direct product $M \oplus N$	Direct sum $M \oplus N$ (Biproduct: $\prod \cong \coprod$)
Poset $(P, \leq)$	Greatest lower bound $\inf(a, b) = a \wedge b$	Least upper bound $\sup(a, b) = a \vee b$

Table 3: Comparison of products and coproducts across algebra and algebraic topology.

build complex spaces by *gluing* simple pieces along common seams. A cylinder is formed by gluing the edges of a rectangle; a sphere is formed by gluing two hemispherical disks along an equator circle ($S^1 \hookrightarrow D^2$); a CW complex is formed by attaching disks along boundary spheres ($S^{k-1} \to X$). A **pushout** is the categorical formalization of gluing along a shared boundary span $A \hookleftarrow C \hookrightarrow B$. Dually, a **pullback** formalizes fiber products and fiber bundles over a base space.

5.3 Pullbacks, Pushouts, and the Seifert–van Kampen Theorem

Often in topology, we do not glue two spaces A and B freely, but rather *along a common subspace* C. This leads to **pullbacks** (gluing over a base) and **pushouts** (gluing along a common source) [18, 3, 6].

Definition 5.4 (Pullback and Pushout [2, 4])**.** 1. Given a cospan $A \xrightarrow{f} C \xleftarrow{g} B$ in $\mathcal{C}$, a **pullback** (or **fiber product**) is an object $P = A \times_C B$ equipped with morphisms $p_1 : P \to A$ and $p_2 : P \to B$ such that $f \circ p_1 = g \circ p_2$, universal among all such commutative squares: for any object Z and morphisms $q_1 : Z \to A$, $q_2 : Z \to B$ with $f \circ q_1 = g \circ q_2$, there exists a unique morphism $u : Z \to P$ such that $p_1 \circ u = q_1$ and $p_2 \circ u = q_2$.

2. Dually, given a span $A \xleftarrow{i} C \xrightarrow{j} B$ in $\mathcal{C}$, a **pushout** (or **amalgamated sum**) is an object $Q = A \amalg_C B$ equipped with morphisms $k_1 : A \to Q$ and $k_2 : B \to Q$ such that $k_1 \circ i = k_2 \circ j$, universal among all such commutative squares: for any object W and morphisms $h_1 : A \to W$, $h_2 : B \to W$ with $h_1 \circ i = h_2 \circ j$, there exists a unique morphism $w : Q \to W$ such that $w \circ k_1 = h_1$ and $w \circ k_2 = h_2$.

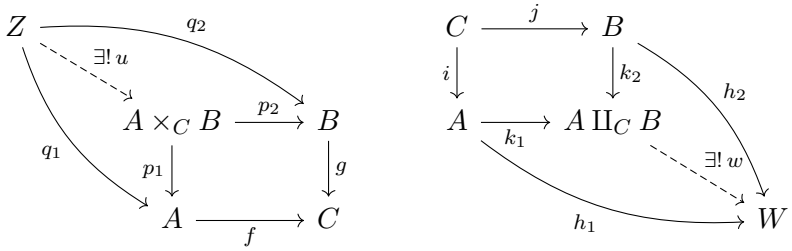


Figure 36: Universal commutative squares for a Pullback $A \times_C B$ (left) and a Pushout $A \amalg_C B$ (right).

Diagram Explanation: In Figure 36 (left), the pullback $A \times_C B$ in **Set** or **Top** is the subspace $\{(a, b) \in A \times B : f(a) = g(b)\}$ (when $g : E \to B$ is a fiber bundle or covering space and $f : A \to B$ is a continuous map, $A \times_B E$ is the **pullback bundle** over A). In Figure 36 (right), the pushout $A \amalg_C B$ in **Top** is the quotient space $(A \sqcup B)/\sim$ where $i(c) \sim j(c)$ for all $c \in C$ (including attaching a k-cell D^k along its boundary $S^{k-1} \to X$ to build a CW complex!).

Remark 5.5 (General Limits and Colimits as Cones over Diagrams [2, 3]). All the constructions in this section are special cases of a single unifying concept. Let $\mathcal{J}$ be a small indexing category and $D : \mathcal{J} \to \mathcal{C}$ a functor (called a **diagram of shape** $\mathcal{J}$). A **limit** $\varprojlim_{\mathcal{J}} D$ is a terminal cone over D, and a **colimit** $\varinjlim_{\mathcal{J}} D$ is an initial cocone under D:

- When $\mathcal{J} = \varnothing$ (empty category), $\varprojlim D$ is a terminal object and $\varinjlim D$ is an initial object.
- When $\mathcal{J} = \{\bullet \quad \bullet\}$ (two discrete objects), $\varprojlim D$ is the product $A \times B$ and $\varinjlim D$ is the coproduct $A \amalg B$.
- When $\mathcal{J} = \{\bullet \to \bullet \leftarrow \bullet\}$, $\varprojlim D$ is the pullback; when $\mathcal{J} = \{\bullet \leftarrow \bullet \to \bullet\}$, $\varinjlim D$ is the pushout.

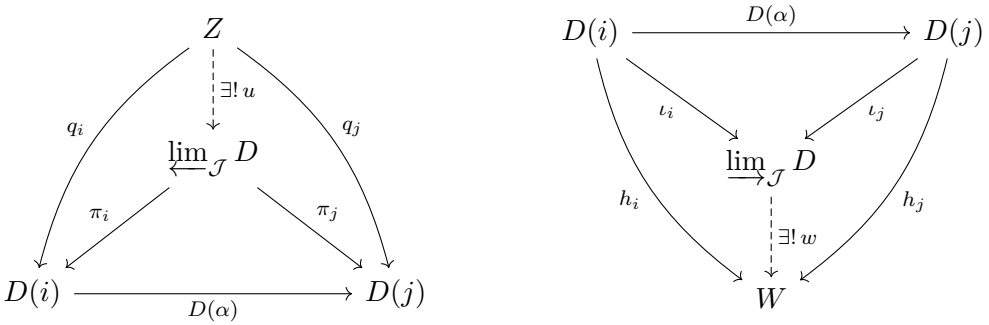


Figure 37: Universal terminal cone for a general Limit $\varprojlim D$ (left) and initial cocone for a Colimit $\varinjlim D$ (right).

Diagram Explanation: In Figure 37, a **cone** from summit Z over the base diagram D consists of leg morphisms $q_i : Z \to D(i)$ commuting with all transition arrows $D(\alpha)$. The **limit** $\varprojlim D$ is the universal (terminal) cone through which every other cone factors uniquely via u. Dually, the **colimit** $\varinjlim D$ is the universal (initial) cocone under D factoring uniquely via w.

5.4 Worked Example Problems and Practice Exercises for Section 5

Worked Example 5.6 (Universal Property of the Product Topology in **Top** [4, 6])**. Question:** Let (X, τ_X) and (Y, τ_Y) be topological spaces, and equip their Cartesian product $X \times Y$ with the standard product topology (generated by basis rectangles $U \times V$ with $U \in \tau_X, V \in \tau_Y$). Prove that $(X \times Y, \pi_1, \pi_2)$ satisfies Definition 5.3(1) in **Top**, and explain why the box topology on $\prod_{n=1}^{\infty} X_n$ fails to be the categorical product for infinite families.

Detailed Solution and Deduction: First, for any open $U \in \tau_X$, $\pi_1^{-1}(U) = U \times Y$ is a basis open set in $X \times Y$, so π_1 (and similarly π_2) is continuous. Now let Z be any topological space and let $f : Z \to X$ and $g : Z \to Y$ be continuous maps. In **Set**, there is already a unique function $h = \langle f, g \rangle : Z \to X \times Y$ satisfying $\pi_1 \circ h = f$ and $\pi_2 \circ h = g$, namely $h(z) = (f(z), g(z))$. We must show that h is continuous with respect to the product topology on $X \times Y$. It suffices to check that the preimage under h of every basis rectangle $U \times V$ $(U \in \tau_X, V \in \tau_Y)$ is open in Z:

$$h^{-1}(U \times V) = \{z \in Z : f(z) \in U \text{ and } g(z) \in V\} = f^{-1}(U) \cap g^{-1}(V).$$

Since f and g are continuous, $f^{-1}(U)$ and $g^{-1}(V)$ are open in Z, and the intersection of two open sets is open in Z. Hence h is continuous! (For a countably infinite product $\prod_{n=1}^{\infty} \mathbb{R}$, the diagonal map $\Delta : \mathbb{R} \to \prod_{n=1}^{\infty} \mathbb{R}$, $t \mapsto (t, t, t, \dots)$, has continuous coordinate projections $\pi_n \circ \Delta = \mathrm{id}_{\mathbb{R}}$, yet $\Delta^{-1}(\prod_{n=1}^{\infty}(-1/n, 1/n)) = \{0\}$ is not open in $\mathbb{R}$, so the box topology violates the universal property of the categorical product!) $\square$

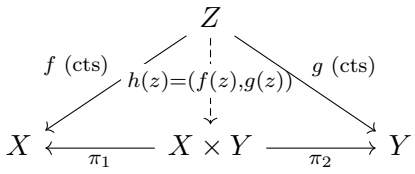


Figure 38: The product topology on $X \times Y$ is the coarsest topology making π_1, π_2 continuous, ensuring h is continuous.

Diagram Explanation: Figure 38 reveals *why* Tychonoff's product topology is defined using finite intersections of cylinder sets $\pi_i^{-1}(U_i)$: it is the unique topology on the Cartesian product that satisfies the universal property of the categorical product in **Top**.

Worked Example 5.7 (The Seifert–van Kampen Theorem as Preservation of Pushouts [6, 7, 20])**.** **Question:** Suppose a topological space $X = U \cup V$ is the union of two path-connected open subsets $U, V \subseteq X$ whose intersection $U \cap V$ is also path-connected and contains the basepoint $x_0 \in U \cap V$. (a) Show that the square of inclusions of $(U \cap V, x_0)$ into (U, x_0) and (V, x_0) and into (X, x_0) is a **pushout square** in $\mathbf{Top}_*$. (b) State the Seifert–van Kampen Theorem in purely categorical terms using the functor $\pi_1 : \mathbf{Top}_* \to \mathbf{Grp}$, and deduce $\pi_1(S^1 \vee S^1, x_0)$ and $\pi_1(S^n, x_0)$ for $n \geq 2$.

Detailed Solution and Deduction:

1. **Part (a) — Pushout in $\mathbf{Top}_*$:** Let W be any pointed space and let $h_1 : (U, x_0) \to (W, w_0)$ and $h_2 : (V, x_0) \to (W, w_0)$ be continuous pointed maps that agree on the intersection: $h_1|_{U\cap V} = h_2|_{U\cap V}$. By the Pasting Lemma of general topology (since U and V are open in X), the piecewise function $w : X \to W$ defined by $w(x) = h_1(x)$ for $x \in U$ and $w(x) = h_2(x)$ for $x \in V$ is well-defined, unique, and continuous. Thus $X = U \amalg_{U\cap V} V$ is a pushout in $\mathbf{Top}_*$.
2. **Part (b) — Categorical Statement & Deductions:** The classical Seifert–van Kampen Theorem states precisely that *the fundamental group functor* $\pi_1 : \mathbf{Top}_* \to \mathbf{Grp}$ *sends open path-connected pushout squares in* $\mathbf{Top}_*$ *to pushout squares (amalgamated free products) in* $\mathbf{Grp}$:
$$\pi_1(U \cup V, x_0) \;\cong\; \pi_1(U, x_0) *_{\pi_1(U\cap V, x_0)} \pi_1(V, x_0).$$
   - For the figure-eight space $X = S^1 \vee S^1$ (formed by gluing two circles at x_0), taking thickened open neighborhoods U, V of each circle that deformation retract onto S^1 with contractible intersection $U \cap V \simeq \{x_0\}$ gives $\pi_1(U \cap V) = \{e\}$, the initial object in $\mathbf{Grp}$. A pushout over the initial object is simply the **coproduct** in $\mathbf{Grp}$ (free product), so $\pi_1(S^1 \vee S^1, x_0) \cong \mathbb{Z} * \mathbb{Z}$ (the free group on 2 generators)!
   - For the n-sphere S^n $(n \geq 2)$, covering S^n by two open hemispheres $U, V \cong \mathbb{R}^n$ (both contractible, so $\pi_1(U) = \pi_1(V) = \{e\}$) whose intersection $U \cap V \simeq S^{n-1}$ is path-connected for $n \geq 2$ gives $\pi_1(S^n, x_0) \cong \{e\} *_{\pi_1(S^{n-1})} \{e\} \cong \{e\}$ (simply connected)! □

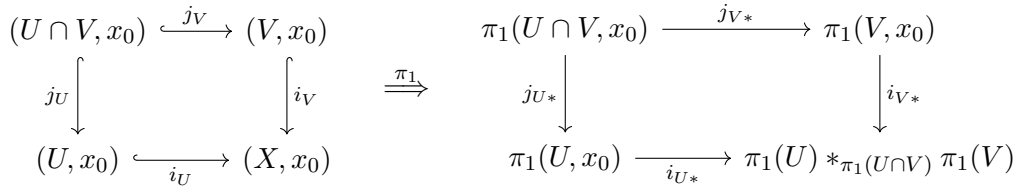


Figure 39: The Seifert–van Kampen Theorem: π_1 preserves open path-connected pushout squares.

Diagram Explanation: Figure 39 crystallizes why category theory revolutionized Algebraic Topology: a complicated theorem about generators and relations of fundamental groups reduces to the single conceptual statement that π_1 transforms a topological pushout square (left) into a group-theoretic pushout square (right).

Practice Exercise 5.1 (Pullback of a Monomorphism is a Monomorphism [3])**.** In Figure 36 (left), suppose $g : B \hookrightarrow C$ is a monomorphism in $\mathcal{C}$. Prove using only the universal property of the pullback $A \times_C B$ that the pulled-back arrow $p_1 : A \times_C B \to A$ is also a monomorphism.

Practice Exercise 5.2 (Equalizers and Kernels)**.** Given parallel morphisms $f, g : A \rightrightarrows B$ in $\mathcal{C}$, their **equalizer** is an object E with a morphism $e : E \to A$ such that $f \circ e = g \circ e$, universal among all morphisms $h : Z \to A$ satisfying $f \circ h = g \circ h$. (a) Draw the universal diagram of an equalizer and prove that $e : E \to A$ is always a monomorphism. (b) In **Ab**, show that the equalizer of a homomorphism $f : A \to B$ and the zero map $0 : A \to B$ is the kernel $\ker(f) \hookrightarrow A$.

Conceptual Bridge to Section 6: The Duality of Adjointness

Throughout Section 5, we proved case-by-case that certain functors preserve products, others preserve pullbacks, and π_1 preserves pushouts. Proving limit and colimit preservation by hand for every functor is laborious. Is there an overarching structural relationship between functors that automates these preservation properties once and for all? In Section 6, we introduce **Adjunctions** ($F \dashv G$). We will prove the celebrated RAPL/LAPC theorem: *right adjoints unconditionally preserve all limits, and left adjoints unconditionally preserve all colimits.* Furthermore, we will prove the crown jewel of category theory—the **Yoneda Lemma**—which guarantees that every object is completely determined by the universe of arrows entering or leaving it.

6 Adjunctions, Representable Functors, and the Yoneda Lemma

Saunders Mac Lane famously remarked in *Categories for the Working Mathematician* [2]: *"Adjoint functors arise everywhere."* Alongside the **Yoneda Lemma**, adjunctions form the crown jewels of foundational category theory [19, 17, 3, 4].

6.1 Adjoint Functors

Definition 6.1 (Adjunction [2, 19, 3])**.** Let $\mathcal{C}$ and $\mathcal{D}$ be locally small categories, and let $F : \mathcal{C} \to \mathcal{D}$ and $G : \mathcal{D} \to \mathcal{C}$ be covariant functors. We say that F is **left adjoint** to G, and G is **right adjoint** to F (written $F \dashv G$), if for every object $X \in \mathrm{Ob}(\mathcal{C})$ and every object $Y \in \mathrm{Ob}(\mathcal{D})$, there is a bijection of hom-sets:

$$\Phi_{X,Y} : \mathrm{Hom}_{\mathcal{D}}(F(X), Y) \xrightarrow{\sim} \mathrm{Hom}_{\mathcal{C}}(X, G(Y)), \tag{11}$$

which is **natural in both** $X \in \mathcal{C}$ **and** $Y \in \mathcal{D}$. Specifically, for any morphisms $f : X' \to X$ in $\mathcal{C}$ and $g : Y \to Y'$ in $\mathcal{D}$, and any $\varphi : F(X) \to Y$ in $\mathcal{D}$, the transpose $\Phi(\varphi) : X \to G(Y)$ satisfies:

$$\Phi_{X',Y'}(g \circ \varphi \circ F(f)) = G(g) \circ \Phi_{X,Y}(\varphi) \circ f. \tag{12}$$

$$\begin{array}{ccc} \mathrm{Hom}_{\mathcal{D}}(F(X), Y) & \xrightarrow[\sim]{\Phi_{X,Y}} & \mathrm{Hom}_{\mathcal{C}}(X, G(Y)) \\ \Big\downarrow{\scriptstyle g\circ(-)\circ F(f)} & & \Big\downarrow{\scriptstyle G(g)\circ(-)\circ f} \\ \mathrm{Hom}_{\mathcal{D}}(F(X'), Y') & \xrightarrow[\Phi_{X',Y'}]{\sim} & \mathrm{Hom}_{\mathcal{C}}(X', G(Y')) \end{array}$$

Figure 40: Naturality square for the hom-set bijection $\Phi_{X,Y}$ of an adjunction $F \dashv G$.

Diagram Explanation: In Figure 40, an adjunction $F \dashv G$ asserts that morphisms out of $F(X)$ into Y in $\mathcal{D}$ correspond one-to-one with morphisms out of X into $G(Y)$ in $\mathcal{C}$. Setting $Y = F(X)$ and applying $\Phi_{X,F(X)}$ to $\mathrm{id}_{F(X)} : F(X) \to F(X)$ yields the **unit of the adjunction** $\eta_X = \Phi(\mathrm{id}_{F(X)}) : X \to G(F(X))$. Dually, setting $X = G(Y)$ and applying Φ^{-1} to $\mathrm{id}_{G(Y)}$ yields the **counit** $\varepsilon_Y : F(G(Y)) \to Y$.

Why Left Adjoints Preserve Colimits (LAPC) and Right Adjoints Preserve Limits (RAPL): In the adjunction bijection $\mathrm{Hom}_{\mathcal{D}}(F(X), Y) \cong \mathrm{Hom}_{\mathcal{C}}(X, G(Y))$, notice where each functor sits: the left adjoint $F(X)$ sits in the **source (left) slot** of $\mathrm{Hom}_{\mathcal{D}}(-, Y)$, which is the slot that defines *colimits* (mapping *out* into test objects Y). Conversely, the right adjoint $G(Y)$ sits in the **target (right) slot** of $\mathrm{Hom}_{\mathcal{C}}(X, -)$, which is the slot that defines *limits* (mapping *in* from test objects X). This placement immediately dictates their preservation laws:

Theorem 6.2 (Right Adjoints Preserve Limits (RAPL) and Left Adjoints Preserve Colimits (LAPC) [2, 3])**.** *Let $F : \mathcal{C} \to \mathcal{D}$ and $G : \mathcal{D} \to \mathcal{C}$ be an adjoint pair $F \dashv G$.*

1. *The right adjoint G preserves all limits that exist in $\mathcal{D}$ (for instance, $G(Y_1 \times Y_2) \cong G(Y_1) \times G(Y_2)$ and $G(T_{\mathcal{D}}) \cong T_{\mathcal{C}}$).*
2. *Dually, the left adjoint F preserves all colimits that exist in $\mathcal{C}$ (for instance, $F(X_1 \amalg X_2) \cong F(X_1) \amalg F(X_2)$ and $F(I_{\mathcal{C}}) \cong I_{\mathcal{D}}$).*

$$\begin{array}{ccc} \mathrm{Hom}_{\mathcal{D}}(F(X_1 \amalg X_2), Y) & \xrightarrow[\sim]{\Phi} & \mathrm{Hom}_{\mathcal{C}}(X_1 \amalg X_2, G(Y)) \\ \sim\uparrow & & \sim\downarrow \text{Univ. Prop. of } \amalg \\ \mathrm{Hom}_{\mathcal{D}}(F(X_1) \amalg F(X_2), Y) & \xleftarrow[\Phi^{-1} \times \Phi^{-1}]{\sim} & \mathrm{Hom}_{\mathcal{C}}(X_1, G(Y)) \times \mathrm{Hom}_{\mathcal{C}}(X_2, G(Y)) \end{array}$$

Figure 41: Chain of natural bijections proving that a left adjoint F preserves coproducts: $F(X_1 \amalg X_2) \cong F(X_1) \amalg F(X_2)$.

Diagram Explanation: Figure 41 shows why Theorem 6.2 has a three-line proof: mapping out of $F(X_1 \amalg X_2)$ into any $Y \in \mathrm{Ob}(\mathcal{D})$ is naturally bijective (via Φ) to mapping out of $X_1 \amalg X_2$ into $G(Y)$, which by the universal property of the coproduct is a pair of maps $(X_1 \to G(Y), X_2 \to G(Y))$, which via Φ^{-1} is a pair of maps $(F(X_1) \to Y, F(X_2) \to Y)$, which is a single map $F(X_1) \amalg F(X_2) \to Y$!

Proof. To see why F preserves coproducts (the proof for arbitrary colimits and limits is identical), let $X_1, X_2 \in \mathrm{Ob}(\mathcal{C})$ have coproduct $X_1 \amalg X_2$. For any test object $Y \in \mathrm{Ob}(\mathcal{D})$, we have the chain of natural bijections displayed in Figure 41:

$$\begin{aligned} \mathrm{Hom}_{\mathcal{D}}(F(X_1 \amalg X_2), Y) &\cong \mathrm{Hom}_{\mathcal{C}}(X_1 \amalg X_2, G(Y)) \\ &\cong \mathrm{Hom}_{\mathcal{C}}(X_1, G(Y)) \times \mathrm{Hom}_{\mathcal{C}}(X_2, G(Y)) \\ &\cong \mathrm{Hom}_{\mathcal{D}}(F(X_1), Y) \times \mathrm{Hom}_{\mathcal{D}}(F(X_2), Y). \end{aligned}$$

This states precisely that $F(X_1 \amalg X_2)$ satisfies the universal property of the coproduct $F(X_1) \amalg F(X_2)$ in $\mathcal{D}$ (Definition 5.3(2)), so $F(X_1 \amalg X_2) \cong F(X_1) \amalg F(X_2)$. Dually, applying the same argument in $\mathcal{C}^{\mathrm{op}}$ and $\mathcal{D}^{\mathrm{op}}$ proves that G preserves all limits. □

The Deep Philosophy of Morphism Probing: Adjunctions demonstrate that an object $F(X)$ in $\mathcal{D}$ can be completely understood by how X maps into $G(Y)$ in $\mathcal{C}$. This exposes a profound philosophical truth: *in category theory, an object possesses no intrinsic meaning in isolation; its entire mathematical identity is encoded in its relationships with all other objects in its universe.* The **Yoneda Lemma** makes this principle mathematically rigorous: it proves that the mapping functor $\mathrm{Hom}_{\mathcal{C}}(A, -)$ contains 100% of the structural information of A. To know an object is to know how every other object sees it.

6.2 The Yoneda Lemma: The Fundamental Theorem of Category Theory

Theorem 6.3 (The Yoneda Lemma [17, 3, 4])**.** *Let $\mathcal{C}$ be a locally small category, let $A \in \mathrm{Ob}(\mathcal{C})$ be an object, and let $F : \mathcal{C} \to \mathbf{Set}$ be any covariant functor. There is a canonical bijection:*

$$\Psi_{A,F} : \mathrm{Hom}_{[\mathcal{C},\mathbf{Set}]}\big(\mathrm{Hom}_{\mathcal{C}}(A,-), F\big) \xrightarrow{\sim} F(A), \qquad \eta \longmapsto \eta_A(\mathrm{id}_A), \tag{13}$$

whose inverse sends an element $x \in F(A)$ to the natural transformation $\eta^{(x)} : \mathrm{Hom}_{\mathcal{C}}(A,-) \Rightarrow F$ with components $\eta^{(x)}_X(f) := F(f)(x)$ for every $f \in \mathrm{Hom}_{\mathcal{C}}(A,X)$. Moreover, $\Psi_{A,F}$ is natural in both $A \in \mathcal{C}$ and $F \in [\mathcal{C},\mathbf{Set}]$.

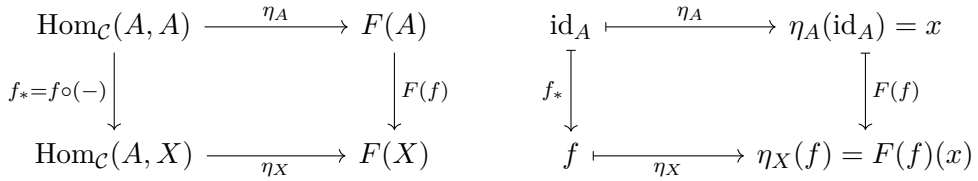


Figure 42: The Yoneda Lemma: chasing $\mathrm{id}_A \in \mathrm{Hom}_{\mathcal{C}}(A,A)$ around the naturality square determines $\eta_X(f)$ for all f.

Diagram Explanation: Look closely at the element chase on the right side of Figure 42. Because $\eta : \mathrm{Hom}_{\mathcal{C}}(A,-) \Rightarrow F$ is a natural transformation, the left square must commute for every morphism $f : A \to X$. Starting with the special element $\mathrm{id}_A \in \mathrm{Hom}_{\mathcal{C}}(A,A)$ in the top-left corner:

- Going **down then right** sends $\mathrm{id}_A \overset{f_*}{\longmapsto} f \circ \mathrm{id}_A = f \overset{\eta_X}{\longmapsto} \eta_X(f)$.
- Going **right then down** sends $\mathrm{id}_A \overset{\eta_A}{\longmapsto} \eta_A(\mathrm{id}_A) \overset{F(f)}{\longmapsto} F(f)\big(\eta_A(\mathrm{id}_A)\big)$.

Equating the two paths forces $\eta_X(f) = F(f)\big(\eta_A(\mathrm{id}_A)\big)$ for *every* object X and *every* arrow $f : A \to X$! Thus the entire natural transformation η is 100% determined by the single element $x = \eta_A(\mathrm{id}_A) \in F(A)$!

Proof. The diagram chase in Figure 42 proves that for any natural transformation $\eta : \mathrm{Hom}_{\mathcal{C}}(A,-) \Rightarrow F$, $\eta_X(f) = F(f)(\eta_A(\mathrm{id}_A))$, so $\Psi_{A,F}$ is injective. Conversely, given any $x \in F(A)$, define $\eta^{(x)}_X(f) := F(f)(x)$. For any morphism $g : X \to Y$ in $\mathcal{C}$ and any $f \in \mathrm{Hom}_{\mathcal{C}}(A,X)$, functoriality of F (Equation (7)) gives:

$$\eta^{(x)}_Y(g_*(f)) = \eta^{(x)}_Y(g \circ f) = F(g \circ f)(x) = (F(g) \circ F(f))(x) = F(g)\big(\eta^{(x)}_X(f)\big),$$

so $\eta^{(x)}$ is indeed a natural transformation, and $\Psi_{A,F}(\eta^{(x)}) = \eta^{(x)}_A(\mathrm{id}_A) = F(\mathrm{id}_A)(x) = \mathrm{id}_{F(A)}(x) = x$. □

Corollary 6.4 (The Yoneda Embedding and Universal Elements [3])**.** *For any objects $A, B \in \mathrm{Ob}(\mathcal{C})$, setting $F = \mathrm{Hom}_{\mathcal{C}}(B,-)$ in Theorem 6.3 yields a bijection:*

$$\mathrm{Hom}_{[\mathcal{C},\mathbf{Set}]}\big(\mathrm{Hom}_{\mathcal{C}}(A,-), \mathrm{Hom}_{\mathcal{C}}(B,-)\big) \;\cong\; \mathrm{Hom}_{\mathcal{C}}(B,A).$$

In particular, $\mathrm{Hom}_{\mathcal{C}}(A,-) \cong \mathrm{Hom}_{\mathcal{C}}(B,-)$ in $[\mathcal{C},\mathbf{Set}]$ if and only if $A \cong B$ in $\mathcal{C}$! An object A is completely determined up to unique isomorphism by the functor of morphisms out of (or into) A.

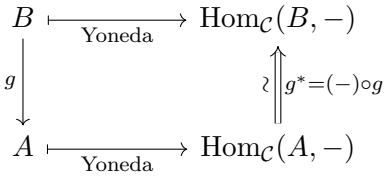


Figure 43: The contravariant Yoneda embedding $\mathcal{C}^{\mathrm{op}} \hookrightarrow [\mathcal{C},\mathbf{Set}]$ is fully faithful: arrows $B \to A$ correspond bijectively to natural transformations $\mathrm{Hom}_{\mathcal{C}}(A,-) \Rightarrow \mathrm{Hom}_{\mathcal{C}}(B,-)$.

Diagram Explanation: Figure 43 illustrates Corollary 6.4: any natural isomorphism between the representable functors $\mathrm{Hom}_{\mathcal{C}}(A,-)$ and $\mathrm{Hom}_{\mathcal{C}}(B,-)$ must be induced by a unique underlying isomorphism $g : B \xrightarrow{\sim} A$ in $\mathcal{C}$. In algebraic topology, this guarantees that spaces representing cohomology (Eilenberg–Mac Lane spaces $K(G,n)$ in Exercise 6.2) are uniquely determined up to homotopy equivalence!

The Loop–Suspension Engine in Homotopy Theory: In algebraic topology, computing the higher homotopy groups $\pi_n(Y) = [S^n, Y]_*$ is notoriously difficult because spheres S^n are geometrically intricate for high n. The loop–suspension adjunction $\Sigma \dashv \Omega$ provides an algebraic crane: because reduced suspension Σ is left adjoint to the loop space functor Ω, we can trade spatial suspension on the source for loop space formation on the target:

$$[S^n, Y]_* \;=\; [\Sigma S^{n-1}, Y]_* \;\cong\; [S^{n-1}, \Omega Y]_*.$$

By iterating this adjunction, computing the n-th homotopy group of an unknown space Y reduces to computing the fundamental group of its $(n-1)$-fold loop space: $\pi_n(Y) \cong \pi_1(\Omega^{n-1}Y)$!

6.3 Worked Example Problems and Practice Exercises for Section 6

Worked Example 6.5 (The Loop–Suspension Adjunction $\Sigma \dashv \Omega$ in $\mathbf{Top}_*$ and $\mathbf{hTop}_*$ [6, 7, 3])**. Question:** For a pointed space (X, x_0), its **reduced suspension** is the quotient space $\Sigma X = (X \times [0,1])/((X \times \{0,1\}) \cup (\{x_0\} \times [0,1]))$, and for (Y, y_0), its **based loop space** is $\Omega Y = \{\gamma : (S^1, s_0) \to (Y, y_0) \text{ continuous}\}$ equipped with the compact-open topology. (a) Explain why Σ is left adjoint to Ω ($\Sigma \dashv \Omega$), inducing a natural bijection $[\Sigma X, Y]_* \cong [X, \Omega Y]_*$ in $\mathbf{hTop}_*$. (b) Deduce that for every $n \geq 1$, the higher homotopy group $\pi_n(Y, y_0) := [(S^n, s_0), (Y, y_0)]_*$ satisfies $\pi_n(Y, y_0) \cong \pi_{n-1}(\Omega Y, c_{y_0}) \cong \cdots \cong \pi_0(\Omega^n Y)$, and that $\Sigma(X_1 \vee X_2) \cong \Sigma X_1 \vee \Sigma X_2$.

Detailed Solution and Deduction:

1. **Part (a) — The Adjunction $\Sigma \dashv \Omega$:** Since $S^1 \cong [0,1]/\{0,1\}$, the reduced suspension is the smash product $\Sigma X \cong X \wedge S^1$. A basepoint-preserving continuous map $\varphi : \Sigma X \to Y$ is a continuous map $X \times [0,1] \to Y$ sending $X \times \{0,1\}$ and $\{x_0\} \times [0,1]$ to y_0. By the exponential law of topology (currying), specifying $\varphi(x,t)$ is equivalent to specifying for each $x \in X$ a based loop $\Phi(\varphi)(x) : S^1 \to Y$ given by $t \mapsto \varphi(x,t)$, with x_0 mapping to the constant loop c_{y_0}. Passing to homotopy classes yields the natural bijection:

$$\mathrm{Hom}_{\mathbf{hTop}_*}(\Sigma X, Y) = [\Sigma X, Y]_* \xrightarrow{\sim} [X, \Omega Y]_* = \mathrm{Hom}_{\mathbf{hTop}_*}(X, \Omega Y).$$

2. **Part (b) — Higher Homotopy Groups and Wedge Sums:** Since $\Sigma S^{n-1} \cong S^n$, iterating the adjunction bijection gives:

$$\pi_n(Y, y_0) = [S^n, Y]_* = [\Sigma S^{n-1}, Y]_* \cong [S^{n-1}, \Omega Y]_* = \pi_{n-1}(\Omega Y, c_{y_0}) \cong \cdots \cong \pi_1(\Omega^{n-1} Y) \cong \pi_0(\Omega^n Y).$$

   Moreover, since Σ is a **left adjoint**, Theorem 6.2(2) (LAPC) immediately guarantees that Σ preserves the coproduct in $\mathbf{Top}_*$ (which is the wedge sum $\vee$ by Table 3): $\Sigma(X_1 \vee X_2) \cong \Sigma X_1 \vee \Sigma X_2$! □

$$\begin{array}{ccc} \big[\Sigma S^{n-1}, Y\big]_* & \xrightarrow[\sim]{\Phi} & \big[S^{n-1}, \Omega Y\big]_* \\ \| & & \| \\ \pi_n(Y, y_0) & \xrightarrow[\sim]{} & \pi_{n-1}(\Omega Y, c_{y_0}) \end{array}$$

Figure 44: The loop–suspension adjunction $\Sigma \dashv \Omega$ shifts dimension in higher homotopy groups.

Diagram Explanation: Figure 44 shows how the adjunction $\Sigma \dashv \Omega$ reduces n-dimensional spheres mapping into Y to $(n-1)$-dimensional spheres mapping into the loop space ΩY. By the Eckmann–Hilton argument [6], this also explains why $\pi_n(Y, y_0)$ is abelian for all $n \geq 2$!

Worked Example 6.6 (Free–Forgetful Adjunction and Free Groups on Disjoint Unions [2, 4])**.** **Question:** Let $F : \mathbf{Set} \to \mathbf{Grp}$ be the free group functor (sending a set S to the free group $F(S)$ generated by S) and $U : \mathbf{Grp} \to \mathbf{Set}$ be the forgetful functor. (a) State the universal property of $F(S)$ as an adjunction $F \dashv U$. (b) Using Theorem 6.2 (LAPC), prove without any word-reduction calculations that $F(S_1 \sqcup S_2) \cong F(S_1) * F(S_2)$.

Detailed Solution and Deduction: (a) For any set $S \in \mathrm{Ob}(\mathbf{Set})$ and any group $G \in \mathrm{Ob}(\mathbf{Grp})$, every function $f : S \to U(G)$ from the generators S into the underlying set of G extends to a unique group homomorphism $\tilde{f} : F(S) \to G$. This defines a natural bijection $\mathrm{Hom}_{\mathbf{Grp}}(F(S), G) \cong \mathrm{Hom}_{\mathbf{Set}}(S, U(G))$, proving $F \dashv U$. (b) Since $F : \mathbf{Set} \to \mathbf{Grp}$ is a left adjoint, Theorem 6.2(2) guarantees that F preserves coproducts. By Table 3, the coproduct in $\mathbf{Set}$ is the disjoint union $S_1 \sqcup S_2$, and the coproduct in $\mathbf{Grp}$ is the free product $*$. Therefore, $F(S_1 \sqcup S_2) \cong F(S_1) * F(S_2)$ immediately! □

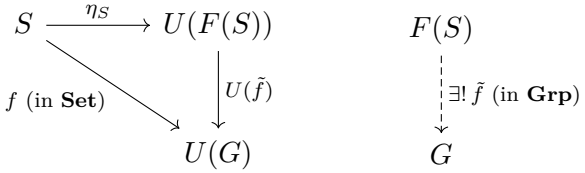


Figure 45: Universal property of the free group $F(S)$ via the unit $\eta_S : S \hookrightarrow U(F(S))$ of $F \dashv U$.

Diagram Explanation: In Figure 45, the inclusion of generators $\eta_S : S \hookrightarrow U(F(S))$ is the unit of the adjunction $F \dashv U$. Any set-map $f : S \to U(G)$ factors uniquely through η_S via a group homomorphism $\tilde{f} : F(S) \to G$.

Practice Exercise 6.1 (Discrete and Indiscrete Topologies as Left and Right Adjoints [3])**.** Let $U : \mathbf{Top} \to \mathbf{Set}$ be the forgetful functor, $D : \mathbf{Set} \to \mathbf{Top}$ be the functor equipping a set with the discrete topology, and $I : \mathbf{Set} \to \mathbf{Top}$ be the functor equipping a set with the indiscrete (trivial) topology. Prove that $D \dashv U \dashv I$, and use Theorem 6.2 to explain why $U : \mathbf{Top} \to \mathbf{Set}$ preserves *both* all limits and all colimits!

Practice Exercise 6.2 (Representability of Cohomology via Eilenberg–Mac Lane Spaces [6])**.** In Algebraic Topology, for an abelian group G and integer $n \geq 1$, an **Eilenberg–Mac Lane space** $K(G, n)$ is a CW complex with $\pi_n(K(G, n)) \cong G$ and $\pi_k(K(G, n)) = 0$ for $k \neq n$. Brown's Representability Theorem shows there is a natural isomorphism of contravariant functors on CW complexes:

$$H^n(X; G) \;\cong\; [X, K(G, n)] = \mathrm{Hom}_{\mathbf{hTop}}(X, K(G, n)).$$

Using the Yoneda Lemma (Theorem 6.3) in $\mathbf{hTop}^{\mathrm{op}}$, explain what natural transformations $H^n(-; G) \Rightarrow H^m(-; G')$ (called **cohomology operations**, such as the Steenrod squares!) correspond to.

7 Where Will You Apply These Concepts? A Roadmap for Algebraic Topology and Beyond

Having mastered the foundational language of categories, functors, natural transformations, limits, colimits, adjunctions, and the Yoneda Lemma, you now hold the master key to modern mathematics. Below is a concrete guide to where you will apply each concept from these lecture notes as you advance in your studies:

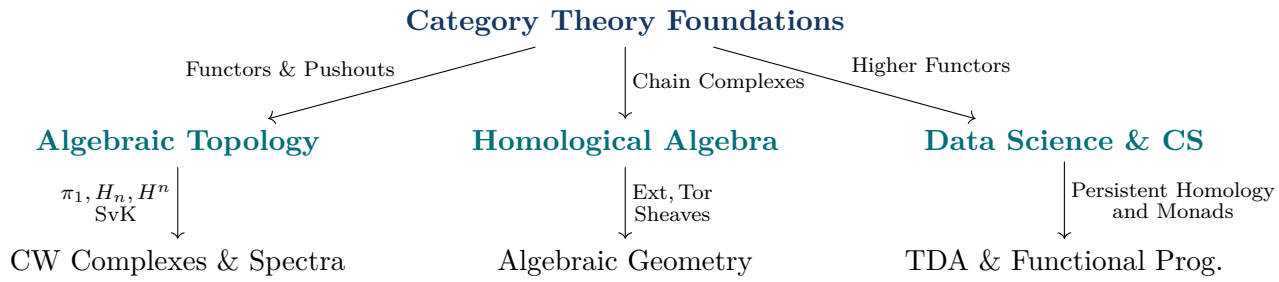


Figure 46: Roadmap of direct applications of foundational category theory across mathematics and science.

Diagram Explanation: Figure 46 maps the concepts developed in Sections 1–6 to three major frontier domains: Algebraic Topology, Homological Algebra/Algebraic Geometry, and Applied Topology/Computer Science.

1. **The Eilenberg–Steenrod Axioms for Homology and Cohomology [6, 7]:** Rather than constructing singular, cellular, and simplicial homology separately from scratch, Samuel Eilenberg and Norman Steenrod showed that *any* sequence of functors $H_n : \mathbf{Top}^{(2)} \to \mathbf{Ab}$ (on pairs of spaces (X, A)) equipped with a natural connecting homomorphism $\partial_n : H_n(X, A) \to H_{n-1}(A)$ satisfying Homotopy, Excision, Dimension, and Additivity (preservation of coproducts!) yields canonically isomorphic homology groups on all CW complexes.

2. **Cellular Homology, CW Complexes, and Gluing via Pushouts [6]:** Every CW complex (spheres, tori, projective spaces $\mathbb{RP}^n, \mathbb{CP}^n$, Grassmannians) is built inductively as an iterated **pushout** (Definition 5.4) of k-disks D^k attached along their boundaries S^{k-1}. Computing $\pi_1(X)$ via Seifert–van Kampen (Worked Example 5.7) or $H_n(X)$ via the Mayer–Vietoris long exact sequence is a direct application of colimits.

3. **Homological Algebra and Derived Functors (Ext and Tor) [2, 10]:** To compute cohomology $H^n(X; G)$ from homology $H_n(X; \mathbb{Z})$, or to compute the homology of a product space $H_n(X \times Y)$, you will use the **Universal Coefficient Theorem** and the **Künneth Formula** [6]. Both rely on the tensor–hom adjunction $(- \otimes_R M) \dashv \mathrm{Hom}_R(M, -)$ and its derived functors $\mathrm{Tor}_1^{\mathbb{Z}}$ and $\mathrm{Ext}^1_{\mathbb{Z}}$.

4. **Topological Data Analysis (TDA), Persistent Homology, and Higher Categories [12, 11]:** In modern data science, a point cloud of high-dimensional data is turned into a filtered sequence of simplicial complexes $K_{r_1} \hookrightarrow K_{r_2} \hookrightarrow \cdots \hookrightarrow K_{r_m}$, which is simply a functor from the poset category $(\mathbb{R}, \leq)$ into **Top**. Applying the homology functor $H_k(-; \mathbb{F})$ produces a **persistence module** $(\mathbb{R}, \leq) \to \mathbf{Vect}_{\mathbb{F}}$, whose barcode decomposition classifies topological features (loops and voids) in real-world biological, sensor, and machine-learning datasets [12].

5. **Relational Databases, Knowledge Representation, and Functional Programming [9]:** As shown by David Spivak [9], a database schema is a small category $\mathcal{S}$, and a database state (instance) is a functor $I : \mathcal{S} \to \mathbf{Set}$. In functional programming (such as Haskell and ML), types and pure functions form a category, polymorphic type constructors (`List`, `Maybe`, `IO`) are endofunctors, and **monads** (arising from adjoint functor pairs $F \dashv G$ via $T = G \circ F$) structure stateful and asynchronous computation.

Navigating the Appendices: Your Foundational Toolkits

To support your transition into advanced coursework and research, these notes conclude with four dedicated reference appendices:

- **Appendix A (Set-Theoretic Foundations):** Resolves Russell's paradox via size stratification, establishing the boundary between small and locally small categories.
- **Appendix B (Abstract Algebra Foundations):** Provides the algebraic spine drawn from coursework notes (**MA204: Abstract Algebra**), bridging group presentations, quotient factorizations, Cayley's theorem, and cohomology coefficient rings.
- **Appendix C (Compendium of Canonical Functors):** A master reference table of 12 primary functors across algebra and geometry, accompanied by verification protocols and non-functorial counterexamples.
- **Appendix D (Hints and Solution Outlines):** Detailed step-by-step guidance for all 12 practice exercises to test and consolidate your understanding.

A Set-Theoretic Foundations: Small vs. Locally Small Categories

In naive set theory, unrestricted comprehension permits forming "the set of all sets", triggering **Russell's Paradox** via the self-referential entity $R = \{S : S \notin S\}$. Because foundational categories such as **Set**, **Grp**, and **Top** take "all sets", "all groups", or "all topological spaces" as objects, category theory requires an explicit foundational stratification to guarantee consistency [2, 4, 5]:

- **Classes vs. Sets (Von Neumann–Bernays–Gödel / Grothendieck Universes):** Under Zermelo–Fraenkel set theory with classes (NBG) or Grothendieck universes $\mathcal{U}$, mathematical collections are stratified into *small sets* (entities whose cardinality is bounded by the universe $\mathcal{U}$) and *proper classes* (collections too vast to belong to $\mathcal{U}$, such as the collection of all sets).
- **Locally Small Categories:** A category $\mathcal{C}$ is **locally small** if for every pair of objects $A, B \in \mathrm{Ob}(\mathcal{C})$, the collection of morphisms $\mathrm{Hom}_{\mathcal{C}}(A, B)$ is a genuine *set* (called a **hom-set**), even though $\mathrm{Ob}(\mathcal{C})$ is a proper class. All standard concrete categories in algebraic topology ($\mathbf{Set}, \mathbf{Grp}, \mathbf{Ab}, \mathbf{Top}, \mathbf{Top}_*, \mathbf{hTop}$) are locally small.
- **Small Categories:** A category $\mathcal{C}$ is **small** if both $\mathrm{Ob}(\mathcal{C})$ and the total collection of all morphisms $\bigcup_{A,B} \mathrm{Hom}_{\mathcal{C}}(A, B)$ are genuine sets (e.g., any finite poset, any group or monoid viewed as a one-object category $\mathcal{B}G$, or the fundamental groupoid $\Pi_1(X)$ of a space X).

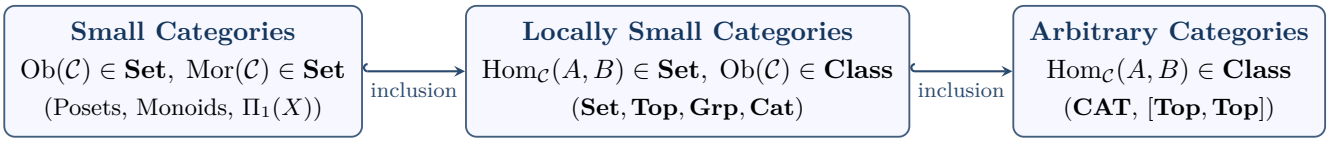


Figure 47: Foundational size hierarchy of categories avoiding Russell's paradox and self-referential incoherence.

Diagram Explanation: Figure 47 stratifies categories into three nested size tiers. A category is **small** if both its objects and arrows form genuine sets; it is **locally small** if arrows between any two objects form a set even if all objects form a proper class; and it is **arbitrary** (strictly large) if its hom-collections can be proper classes (such as **CAT**, the category of all locally small categories). Crucially, **Cat** (the category of all small categories) is locally small and large, ensuring that $\mathbf{Cat} \notin \mathrm{Ob}(\mathbf{Cat})$ and precluding Russell's paradox.

B Abstract Algebra Foundations: The Algebraic Spine of Topology

Coursework Reference: This algebraic bridge draws upon coursework taught by Dr. Gerald Williams at the University of Essex. Complete course slides and problem sets are available on GitHub: `MA204: Abstract Algebra (Sections 1-12)`.

Every topological invariant studied in this monograph translates spatial topology into algebraic data. Below, we review the core algebraic theorems from foundational abstract algebra (MA204) and demonstrate how they translate directly into categorical language.

B.1 Groups, Homomorphisms, and the First Isomorphism Theorem

Recall from MA204 that a **group** $(G, \cdot, e)$ is a set equipped with an associative binary operation, a unique two-sided identity e, and unique inverses g^{-1} for every $g \in G$ (Lemma 3.15 and Lemma 3.16 in MA204).

Definition B.1 (Normal Subgroup and Quotient Group)**.** A subgroup $N \leq G$ is called a **normal subgroup** (denoted $N \trianglelefteq G$) if it is invariant under all inner automorphisms:

$$gng^{-1} \in N \qquad \text{for all } g \in G \text{ and } n \in N.$$

The set of left cosets $G/N = \{gN : g \in G\}$ then forms a group under the operation $(g_1N)(g_2N) = (g_1g_2)N$, equipped with a canonical surjective quotient homomorphism $\pi : G \twoheadrightarrow G/N$.

Theorem B.2 (First Isomorphism Theorem for Groups)**.** *Let $\phi : G \to H$ be a group homomorphism. Then the kernel $\ker(\phi) = \{g \in G : \phi(g) = e_H\}$ is a normal subgroup of G, the image $\mathrm{im}(\phi) = \{\phi(g) : g \in G\}$ is a subgroup of H, and there exists a unique isomorphism $\bar{\phi} : G/\ker(\phi) \xrightarrow{\sim} \mathrm{im}(\phi)$ making the following triangle commute:*

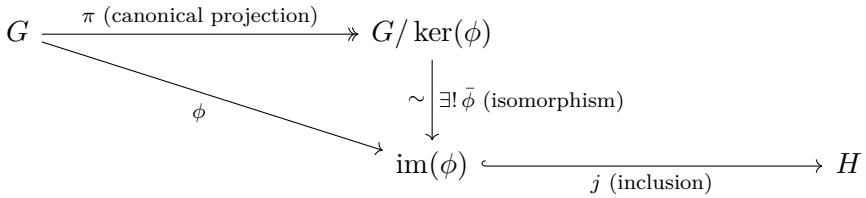


Figure 48: Categorical universal factorization of the First Isomorphism Theorem in **Grp**.

Diagram Explanation: In Figure 48, the First Isomorphism Theorem is cast in its universal arrow-theoretic form: any homomorphism ϕ factors canonically into an epimorphism π followed by a core isomorphism $\bar{\phi}$ followed by a monomorphism j. In singular homology (Definition 3.6), this theorem guarantees that the cycle-boundary quotient $H_n(X) = \ker(\partial_n)/\mathrm{im}(\partial_{n+1})$ is an abelian group.

B.2 Free Groups, Presentations, and Amalgamated Pushouts

In Section 5, Table 3 showed that the categorical product in **Grp** is the direct product $G \times H$ (MA204 Section 8), whereas the categorical coproduct is the **free product** $G * H$.

Definition B.3 (Group Presentation)**.** A group G has presentation $\langle S \mid R \rangle$ (where S is a set of generators and $R \subseteq F(S)$ is a set of relators) if G is isomorphic to the quotient of the free group $F(S)$ by the normal closure $\langle\langle R \rangle\rangle$ of R:

$$G \;\cong\; F(S)/\langle\langle R \rangle\rangle.$$

In Algebraic Topology, calculating the fundamental group $\pi_1(X, x_0)$ via the Seifert–van Kampen Theorem (Worked Example 5.7) reduces topological cell attachment directly to group presentations via categorical pushouts:

Proposition B.4 (Pushouts in **Grp** as Amalgamated Free Products)**.** *Given homomorphisms $\phi_1 : K \to G_1$ and $\phi_2 : K \to G_2$, the categorical pushout $G_1 \amalg_K G_2$ in* **Grp** *is the* ***amalgamated free product****:*

$$G_1 *_K G_2 \;\cong\; (G_1 * G_2)/\langle\langle \phi_1(k)\phi_2(k)^{-1} : k \in K \rangle\rangle.$$

If $G_1 = \langle S_1 \mid R_1 \rangle$ and $G_2 = \langle S_2 \mid R_2 \rangle$, then:

$$G_1 *_K G_2 \;\cong\; \langle S_1 \sqcup S_2 \mid R_1 \cup R_2 \cup \{\phi_1(k) = \phi_2(k) : k \in K\} \rangle.$$

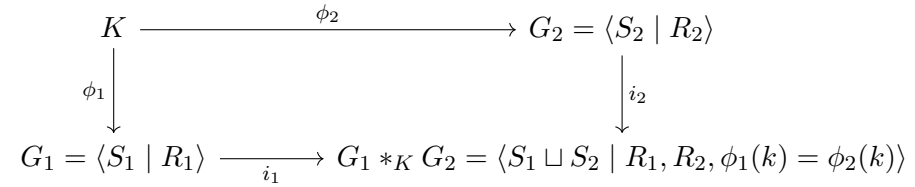


Figure 49: Pushout square in **Grp**: amalgamating generators $S_1 \sqcup S_2$ and identifying shared relators $\phi_1(k) = \phi_2(k)$.

Diagram Explanation: In Figure 49, the pushout in **Grp** first forms the free product $G_1 * G_2$ (taking the disjoint union of generators $S_1 \sqcup S_2$ and relations $R_1 \cup R_2$) and then quotients by the normal subgroup generated by $\phi_1(k)\phi_2(k)^{-1}$ so the two images of K agree inside the amalgam.

B.3 Cayley's Representation Theorem as the Prototypical Yoneda Lemma

In MA204 Section 6, Theorem 6.9 proves that every finite permutation set forms a symmetric group S_n. Arthur Cayley's landmark 1854 theorem asserts that every group is isomorphic to a concrete permutation group. Arrow-theoretically, Cayley's theorem is nothing other than the **one-object specialization of the Yoneda Lemma** (Theorem 6.3):

Theorem B.5 (Cayley's Representation Theorem)**.** *Every group G is isomorphic to a subgroup of the symmetric group* $\mathrm{Sym}(G)$.

$$\mathcal{B}G \text{ (One-Object Category } \bullet\text{)} \xhookrightarrow{\text{Yoneda } h^\bullet} [\mathcal{B}G, \mathbf{Set}] \text{ (Left } G\text{-Sets)}$$

$$g \in \mathrm{Hom}_{\mathcal{B}G}(\bullet,\bullet) = G \xmapsto[\text{Faithful}]{} \lambda_g = g \cdot (-) \in \mathrm{Sym}(G)$$

Figure 50: Cayley's Theorem as the Yoneda embedding applied to a one-object group category $\mathcal{B}G$.

Diagram Explanation: In Figure 50, viewing a group G as the endomorphism group of a single object $\bullet$ in $\mathcal{B}G$, the covariant representable functor $h^\bullet = \mathrm{Hom}_{\mathcal{B}G}(\bullet, -)$ sends $\bullet$ to the underlying set G and sends each group element $g \in G$ to the left-multiplication permutation $\lambda_g(x) = g \cdot x$ in $\mathrm{Sym}(G)$.

Categorical Deduction from the Yoneda Lemma. Regard G as a category $\mathcal{B}G$ with a single object $\bullet$, where $\mathrm{Hom}_{\mathcal{B}G}(\bullet,\bullet) = G$ and morphism composition is the group operation $g \circ h = gh$ (Section 1.1). Consider the covariant representable functor $h^\bullet = \mathrm{Hom}_{\mathcal{B}G}(\bullet, -) : \mathcal{B}G \to \mathbf{Set}$ (Definition 3.7(1)):

1. On the single object $\bullet$, $h^\bullet(\bullet) = \mathrm{Hom}_{\mathcal{B}G}(\bullet,\bullet) = G$.
2. For any element $g \in G = \mathrm{Hom}_{\mathcal{B}G}(\bullet,\bullet)$, the functor $h^\bullet$ assigns the set-map $h^\bullet(g) = g_* : G \to G$ acting by post-composition: $x \mapsto g \cdot x$ (the **left regular representation** λ_g). Because $h^\bullet$ is covariant, $\lambda_{gh}(x) = (gh) \cdot x = g \cdot (h \cdot x) = (\lambda_g \circ \lambda_h)(x)$, so $g \mapsto \lambda_g$ preserves multiplication order directly.

By the Yoneda Embedding (Corollary 6.4 applied to $\mathcal{B}G^{\mathrm{op}}$), the assignment $g \mapsto \lambda_g$ is a fully faithful functor. Fully faithful functors are injective on hom-sets (Definition 3.9), so the group homomorphism $\lambda : G \hookrightarrow \mathrm{Sym}(G)$ is injective, establishing an isomorphism $G \cong \mathrm{im}(\lambda) \leq \mathrm{Sym}(G)$. □

B.4 Rings, Ideals, and Coefficient Fields in Cohomology

In MA204 Sections 10–12, rings $(R, +, \cdot)$, integral domains, and fields $\mathbb{F}$ are classified. In Algebraic Topology, this structure governs the coefficient systems of cohomology:

1. **Cohomology as a Graded Ring:** While singular homology $\bigoplus_{n\geq 0} H_n(X;\mathbb{Z})$ is merely a graded abelian group, the cohomology $H^*(X;R) = \bigoplus_{n\geq 0} H^n(X;R)$ forms a **graded-commutative ring** under the cup product $\smile$: $H^k(X;R) \times H^l(X;R) \to H^{k+l}(X;R)$ satisfying $\alpha \smile \beta = (-1)^{kl}(\beta \smile \alpha)$.

2. **Field Coefficients and Dimension:** When coefficients are taken in a field $\mathbb{F}$ (such as $\mathbb{Q}$ or $\mathbb{Z}_p$ for prime p, MA204 Corollary 12.15), $H_n(X;\mathbb{F})$ and $H^n(X;\mathbb{F})$ become objects in $\mathbf{Vect}_{\mathbb{F}}$. By the Universal Coefficient Theorem for cohomology:

$$H^n(X;\mathbb{F}) \;\cong\; \mathrm{Hom}_{\mathbb{F}}\big(H_n(X;\mathbb{F}),\mathbb{F}\big) \;=\; H_n(X;\mathbb{F})^*,$$

which is the exact dual vector space functor studied in Worked Example 4.5.

C Compendium of Canonical Functors and Functoriality Verification

Throughout pure mathematics and algebraic topology, mathematical structures are connected by canonical functors. Table 4 provides a comprehensive reference of the primary functors encountered in undergraduate and early graduate curricula.

Functor Name	Source → Target	Variance	Object Assignment	Morphism Assignment
Forgetful (U)	$\mathbf{Grp} \to \mathbf{Set}$	Covariant	$(G,\cdot) \mapsto G$	$(f: G \to H) \mapsto f$
Free Group (F)	$\mathbf{Set} \to \mathbf{Grp}$	Covariant	$S \mapsto F(S)$	$(f: S \to T) \mapsto F(f)$
Abelianization	$\mathbf{Grp} \to \mathbf{Ab}$	Covariant	$G \mapsto G/[G,G]$	$f \mapsto f^{\mathrm{ab}}$
Discrete Topology	$\mathbf{Set} \to \mathbf{Top}$	Covariant	$S \mapsto (S, \mathcal{P}(S))$	$f \mapsto f$
Fundamental Group	$\mathbf{Top}_* \to \mathbf{Grp}$	Covariant	$(X,x_0) \mapsto \pi_1(X,x_0)$	$f \mapsto (f_* : [\gamma] \mapsto [f \circ \gamma])$
Higher Homotopy	$\mathbf{Top}_* \to \mathbf{Ab}$	Covariant	$(X,x_0) \mapsto \pi_n(X,x_0)$	$f \mapsto f_* \quad (n \geq 2)$
Singular Homology	$\mathbf{Top} \to \mathbf{Ab}$	Covariant	$X \mapsto H_n(X;\mathbb{Z})$	$f \mapsto (f_* : [z] \mapsto [f_\#(z)])$
Singular Cohomology	$\mathbf{Top}^{\mathrm{op}} \to \mathbf{Ab}$	**Contravariant**	$X \mapsto H^n(X;G)$	$f \mapsto (f^* : [\psi] \mapsto [\psi \circ f_\#])$
Suspension (Σ)	$\mathbf{Top}_* \to \mathbf{Top}_*$	Covariant	$X \mapsto X \wedge S^1$	$f \mapsto f \wedge \mathrm{id}_{S^1}$
Loop Space (Ω)	$\mathbf{Top}_* \to \mathbf{Top}_*$	Covariant	$Y \mapsto \Omega Y$	$g \mapsto (g \circ -)$
Dual Space ($(-)^*$)	$\mathbf{Vect}_k^{\mathrm{op}} \to \mathbf{Vect}_k$	**Contravariant**	$V \mapsto \mathrm{Hom}_k(V,k)$	$(T: V \to W) \mapsto T^* = - \circ T$
Double Dual ($(-)^{**}$)	$\mathbf{Vect}_k \to \mathbf{Vect}_k$	Covariant	$V \mapsto V^{**}$	$T \mapsto T^{**}$

Table 4: Master compendium of canonical functors across algebra, topology, and geometry.

The 4-Step Functoriality Proof Protocol for Students

When encountering or constructing a candidate functor $F : \mathcal{C} \to \mathcal{D}$, verify four independent conditions:

1. **Object Well-Definedness:** Check that for every object $X \in \mathrm{Ob}(\mathcal{C})$, the target entity $F(X)$ satisfies all defining properties of an object in $\mathcal{D}$ (e.g., verifying that $\pi_1(X,x_0)$ is a genuine group under loop concatenation).

2. **Morphism Well-Definedness:** Check that for every arrow $f : X \to Y$ in $\mathcal{C}$, the assigned arrow $F(f)$ genuinely belongs to $\mathrm{Hom}_{\mathcal{D}}(F(X), F(Y))$ (e.g., verifying that $f_*([f \circ \gamma])$ is independent of the choice of representative loop γ).
3. **Identity Preservation:** Verify that $F(\mathrm{id}_X) = \mathrm{id}_{F(X)}$ (Equation (5)).
4. **Composition Preservation:** Verify that $F(g \circ f) = F(g) \circ F(f)$ for covariant functors (or $F(g \circ f) = F(f) \circ F(g)$ for contravariant functors).

The Danger Zone: Classical Non-Functorial Operations

Not every assignment of objects to objects can be extended to a functor. A common beginner error is assuming that every natural geometric or algebraic construction is functorial. Two prominent counterexamples include:

1. The Center of a Group is NOT Functorial on Grp

Assigning to each group G its center $Z(G) = \{z \in G : zg = gz \ \forall g \in G\}$ is well-defined on objects. However, there is **no** functor $Z : \mathbf{Grp} \to \mathbf{Ab}$.

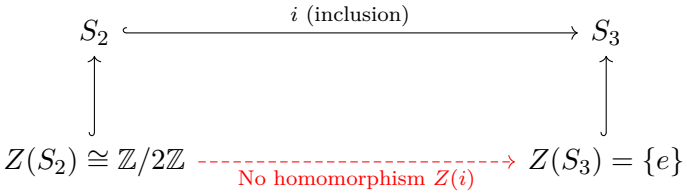


Figure 51: Obstruction to functoriality of the center: the inclusion $S_2 \hookrightarrow S_3$ maps the generator of $Z(S_2)$ to a transposition $(12) \notin Z(S_3)$, preventing the square from completing.

Diagram Explanation: In Figure 51, the inclusion of the two-element symmetric group $S_2 \cong \mathbb{Z}/2\mathbb{Z}$ into S_3 illustrates why the center fails to be functorial. Because S_2 is abelian, $Z(S_2) = S_2$. However, S_3 has a trivial center $Z(S_3) = \{e\}$. The inclusion $i : S_2 \hookrightarrow S_3$ maps the non-trivial generator of $Z(S_2)$ to the transposition $(12) \in S_3$, which does *not* commute with all permutations in S_3. Hence, i cannot restrict to a homomorphism $Z(S_2) \to Z(S_3)$, proving that Z cannot be a functor.

2. The Topological Boundary is NOT Functorial on Top

Assigning to a topological space with boundary X its boundary ∂X is well-defined on objects. However, there is **no** functor $\partial : \mathbf{Top} \to \mathbf{Top}$.

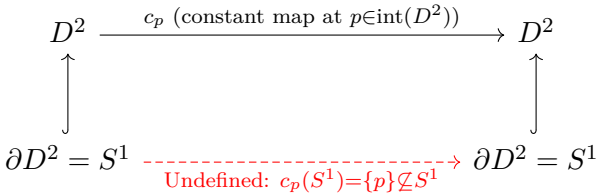


Figure 52: Obstruction to functoriality of the topological boundary: a constant interior map sends the boundary outside itself.

Diagram Explanation: In Figure 52, let D^2 be the closed 2-disk with boundary $\partial D^2 = S^1$. Choose an interior point $p \in \mathrm{int}(D^2)$ and consider the constant continuous map $c_p : D^2 \to D^2$ sending all of D^2 to p. If ∂ were a functor, c_p would induce a continuous map $\partial(c_p) : \partial D^2 \to \partial D^2$. But $c_p(\partial D^2) = \{p\} \not\subseteq S^1$. Hence continuous maps do not preserve boundaries, ruling out functoriality.

D Hints and Solution Outlines for Selected Practice Exercises

- **Hint for Exercise 1.1 (Slice Category $\mathcal{C}/B$):** Given objects (X, p_X), (Y, p_Y), (Z, p_Z) and morphisms $f : (X, p_X) \to (Y, p_Y)$ and $g : (Y, p_Y) \to (Z, p_Z)$, by definition $p_Y \circ f = p_X$ and $p_Z \circ g = p_Y$. Composing in $\mathcal{C}$ gives $p_Z \circ (g \circ f) = (p_Z \circ g) \circ f = p_Y \circ f = p_X$, so $g \circ f$ satisfies the slice compatibility condition. The identity $\mathrm{id}_X : X \to X$ satisfies $p_X \circ \mathrm{id}_X = p_X$, serving as $\mathrm{id}_{(X,p_X)}$. Associativity inherits directly from $\mathcal{C}$.

- **Hint for Exercise 1.2 (Product Category $\mathcal{C} \times \mathcal{D}$):** Objects are pairs (C, D) with $C \in \mathrm{Ob}(\mathcal{C})$ and $D \in \mathrm{Ob}(\mathcal{D})$. Morphisms are pairs $(f, g) : (C, D) \to (C', D')$ with $f \in \mathrm{Hom}_{\mathcal{C}}(C, C')$ and $g \in \mathrm{Hom}_{\mathcal{D}}(D, D')$. Define componentwise composition $(f', g') \circ (f, g) = (f' \circ_{\mathcal{C}} f, g' \circ_{\mathcal{D}} g)$ and identities $\mathrm{id}_{(C,D)} = (\mathrm{id}_C, \mathrm{id}_D)$. Verification of associativity and unitality follows immediately from the coordinates.

- **Hint for Exercise 2.1 (Set Monomorphisms and Epimorphisms):**
  - *Monomorphism $\implies$ Injective:* If $f(a_1) = f(a_2)$ for $a_1 \neq a_2 \in A$, define two constant functions $g_1, g_2 : \{*\} \to A$ by $g_1(*) = a_1$ and $g_2(*) = a_2$. Then $f \circ g_1 = f \circ g_2$. Since f is mono, $g_1 = g_2$, forcing $a_1 = a_2$.
  - *Epimorphism $\implies$ Surjective:* If $f(A) \subsetneq B$, choose $b_0 \in B \setminus f(A)$. Define $h_1, h_2 : B \to \{0, 1\}$ where $h_1(b) = 1$ for all b, and $h_2(b) = 1$ if $b \in f(A)$ while $h_2(b_0) = 0$. Then $h_1 \circ f = h_2 \circ f$ on all of A, but $h_1 \neq h_2$, contradicting right-cancellability.

- **Hint for Exercise 2.2 (Dense Epimorphisms in Haus):** Let $h_1, h_2 : Y \rightrightarrows W$ be continuous maps into a Hausdorff space W such that $h_1 \circ f = h_2 \circ f$. The equalizer subset $E = \{y \in Y : h_1(y) = h_2(y)\}$ is the preimage of the diagonal $\Delta_W \subseteq W \times W$ under the continuous map $(h_1, h_2) : Y \to W \times W$. Because W is Hausdorff, Δ_W is closed, so E is closed in Y. Since E contains the dense image $f(X)$, $E = \overline{E} \supseteq \overline{f(X)} = Y$. Hence $h_1 = h_2$.

- **Hint for Exercise 3.1 (Fully Faithful Functors Reflect Isomorphisms):** If $F(f) : F(A) \to F(B)$ is an isomorphism with inverse $\beta : F(B) \to F(A)$ in $\mathcal{D}$, fullness of F guarantees there exists a morphism $g : B \to A$ in $\mathcal{C}$ such that $F(g) = \beta$. Then $F(g \circ f) = F(g) \circ F(f) = \beta \circ F(f) = \mathrm{id}_{F(A)} = F(\mathrm{id}_A)$. Because F is faithful (injective on hom-sets), $g \circ f = \mathrm{id}_A$. Similarly, $f \circ g = \mathrm{id}_B$. Thus f is an isomorphism in $\mathcal{C}$.

- **Hint for Exercise 3.2 (Abelianization Functor):** For any homomorphism $\phi : G \to H$, $\phi([x, y]) = \phi(xyx^{-1}y^{-1}) = [\phi(x), \phi(y)] \in [H, H]$, so $\phi([G, G]) \subseteq [H, H]$. By the First Isomorphism Theorem (Theorem B.2), ϕ induces a well-defined homomorphism $\phi^{\mathrm{ab}} : G/[G, G] \to H/[H, H]$ by $\phi^{\mathrm{ab}}(g[G, G]) = \phi(g)[H, H]$. Functoriality follows from $(\psi \circ \phi)^{\mathrm{ab}} = \psi^{\mathrm{ab}} \circ \phi^{\mathrm{ab}}$.

- **Hint for Exercise 4.1 (Determinant as a Natural Transformation):** For any unital ring homomorphism $\phi : R \to S$, the induced matrix homomorphism $\mathrm{GL}_n(\phi) : \mathrm{GL}_n(R) \to \mathrm{GL}_n(S)$ applies ϕ entrywise. Using the Leibniz formula $\det(M) = \sum_{\sigma \in S_n} \mathrm{sgn}(\sigma) \prod_{i=1}^{n} M_{i,\sigma(i)}$, applying ϕ distributes across addition and multiplication, proving $\det_S(\mathrm{GL}_n(\phi)(M)) = \phi(\det_R(M))$. Hence the naturality square commutes.

- **Hint for Exercise 4.2 (Equivalence of Skeletons):** The inclusion functor $I : \mathcal{S} \hookrightarrow \mathbf{FinSet}$ is full and faithful by definition because $\mathcal{S}$ is a full subcategory. Every finite set X with $|X| = n \geq 0$ (with $\mathbf{0} = \varnothing$) is in bijection with $\mathbf{n} = \{1, 2, \ldots, n\}$, so $X \cong I(\mathbf{n})$ in **FinSet**. Thus I is essentially surjective, establishing $\mathcal{S} \simeq \mathbf{FinSet}$.

- **Hint for Exercise 5.1 (Pullback of a Monomorphism):** In Figure 36 (left), suppose $g : B \hookrightarrow C$ is mono, and let $u, v : Z \rightrightarrows A \times_C B$ satisfy $p_1 \circ u = p_1 \circ v$. Then $g \circ (p_2 \circ u) = f \circ (p_1 \circ u) = f \circ (p_1 \circ v) = g \circ (p_2 \circ v)$. Because g is left-cancellable, $p_2 \circ u = p_2 \circ v$. By uniqueness of the induced map into the pullback, $u = v$, proving p_1 is a monomorphism.

- **Hint for Exercise 5.2 (Equalizers and Kernels):**
  - (a) Let $u, v : Z \rightrightarrows E$ satisfy $e \circ u = e \circ v$. Since $e : E \to A$ equalizes f and g, $f \circ (e \circ u) = g \circ (e \circ u)$. By the universal property of E, there exists a *unique* map $h : Z \to E$ such that $e \circ h = e \circ u$. Both u and v satisfy this condition, forcing $u = v$.
  - (b) In **Ab**, the condition $f(e(x)) = 0(e(x)) = 0$ means $e(E) \subseteq \ker(f)$. The universal property of $\ker(f)$ coincides with the equalizer universal property.
- **Hint for Exercise 6.1 ($D \dashv U \dashv I$ Adjunction Chain):** Any function from a discrete topological space $D(S)$ to a space X is continuous, establishing a natural bijection $\mathrm{Hom}_{\mathbf{Top}}(D(S), X) \cong \mathrm{Hom}_{\mathbf{Set}}(S, U(X))$, so $D \dashv U$. Conversely, any function from a space X into an indiscrete space $I(S)$ is continuous, giving $\mathrm{Hom}_{\mathbf{Top}}(X, I(S)) \cong \mathrm{Hom}_{\mathbf{Set}}(U(X), S)$, so $U \dashv I$. By Theorem 6.2, since U is both a right adjoint and a left adjoint, it preserves all limits and all colimits.
- **Hint for Exercise 6.2 (Cohomology Operations via Yoneda):** By Brown's Representability Theorem, $H^n(-; G) \cong \mathrm{Hom}_{\mathbf{hTop}}(-, K(G, n))$. In the opposite category $\mathbf{hTop}^{\mathrm{op}}$, this is a covariant representable functor $\mathrm{Hom}_{\mathbf{hTop}^{\mathrm{op}}}(K(G, n), -)$. Applying the Yoneda Lemma (Theorem 6.3) in $\mathbf{hTop}^{\mathrm{op}}$ yields:

  $$\mathrm{Nat}\big(H^n(-; G), H^m(-; G')\big) \;\cong\; \mathrm{Hom}_{\mathbf{hTop}}\big(K(G, n), K(G', m)\big) \;=\; [K(G, n), K(G', m)]_*.$$

  Evaluating on the fundamental class $\iota_n \in H^n(K(G, n); G)$ identifies every natural cohomology operation with a unique cohomology class in $H^m(K(G, n); G')$.

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